Numerical Simulation of Full-Scale Static Load Tests Using PLAXIS 3D: Insights and Applications

by

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Md. Ashrafuzzaman

Dedicated

to

*"**My Parents**"*

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ABSTRACT

Accurately estimating the ultimate bearing capacity (UBC) of soil for a pile is critical for geotechnical engineers to ensure safe and economical deep foundation designs. Traditionally, there were two main approaches: analytical or empirical methods and static load tests (SLTs). Analytical methods employ mathematical equations or empirical correlations derived from soil qualities and pile features. These methods offer simplicity and speed, yet frequently come short in terms of practical reliability due to their over-reliance on simplified assumptions. Static load tests (SLTs) involve applying a controlled load to a test pile, allowing for settlement measurement. This method yields reliable and accurate data that is specific to the site. However, it is a tedious and expensive procedure. Applying PLAXIS 3D for finite element simulations provides an appealing option to estimate ultimate bearing capacity (UBC) alongside the traditional methods, such as empirical equations and in-situ static load testing (SLT). This paper intends to explore and evaluate the accuracy and reliability of a constitutive model simulated using PLAXIS 3D in order to replicate the SLT process. The simulation accurately models soil types and stratigraphy, recreating the surrounding soil and soil-structure interaction using ideal settings. Numerical modelling of static load tests (SLTs) allows a thorough evaluation of stratified soil layers, navigating the limitations of analytical and empirical methods to estimate ultimate bearing capacity (UBC). These simulations provide estimated ultimate bearing capacity in an economical, quick, and reliable manner compared to actual static load tests (SLTs), which require intricate equipment and skilled supervision.

This study utilized authentic data obtained from static load tests (SLTs) and geotechnical investigation reports prepared for the Co-operative Bhaban located in Shahbagh. Regarding TP-4, the simulations replicated the actual test pile, which was a cylindrical concrete pile with a diameter of 558.8 mm (22 inches) and a length of 13 m (43 feet), using characteristics from BH-2 due to their close vicinity.

Various mesh sizes in PLAXIS 3D generated distinct Q-z curves, which were then compared with the Q-z curve obtained from the static load test (SLT) to determine the most appropriate match. The application of a very fine mesh size resulted in the most accurate alignment, leading to an estimated UBC (ultimate bearing capacity) of 183 tonnes according to the Davisson technique (1973). The ultimate bearing capacity

(UBC) closely aligned with the benchmark of 186 tonnes obtained from the static load test (SLT), with a variation of approximately 1.613%.

The reliability of the simulations was established by comparing them with estimated ultimate bearing capacity (UBC) values calculated using the analytical method (O'Neill and Reese method) and empirical correlations (Meyerhof). By employing a conservative approach that incorporates a reduced modulus of elasticity, a lower ultimate bearing capacity (UBC) is obtained, hence ensuring safety by deliberately underestimating the capacity.

In summary, PLAXIS 3D simulations accurately reproduce SLT results, offering a dependable and streamlined approach for estimating UBC in geotechnical engineering.

Keywords

Pile, Static Load Test, Finite Element Method, PLAXIS 3D, Ultimate Bearing Capacity, Davisson Method.

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CHAPTER 1
INTRODUCTION

1.1 Background and Motivations

Pile foundation is an approach with a wide range of preferences and applications for designing the deep foundation to provide needed support for superstructures. (Wei et al., 2020) These structures include high-rise buildings, large highway bridges, harbors, wind power plants, and oil extraction facilities. Their popularity stems from their high load-bearing capacity and extensive application history across various projects. (Wei et al., 2020) & (Ren & Sun, 2023)

However, accurate estimation of the ultimate bearing capacity of piles becomes a challenging job due to having a complex soil-pile interaction. (Naggar, 2004) Several experimental and theoretical methods for estimating pile bearing capacity depend on the simplifying assumptions about the parameters that determine ultimate bearing capacity. (Naggar, 2004) These assumptions simplify the analysis of tests such as the cone penetration test (CPT) (Kordjazi et al., 2014), dynamic load test (DLT) (Heins & Grabe, 2019), standard penetration test (SPT) (Shooshpasha, 2013) & (Hasanzadeh & Shooshpasha, 2017), and static load test (SLT) (Sakr, 2013) under load. Among these methods, the static load test (SLT) (Sakr, 2013) is considered as the most reliable and direct approach. In a static load test, a pile is subjected to a gradually incrementing axial load until failure occurs, providing the most accurate measurement of its ultimate bearing capacity. (Ren & Sun, 2023) Estimation of the ultimate bearing capacity in a rational manner is considered one of the key challenges in pile foundation design. (Birid, 2017)

In projects that involve pile foundations, verifying the actual ultimate bearing capacity against the theoretical ultimate pile capacity is crucial. (Birid, 2017) As it is well known that the accurate estimation of ultimate bearing capacity is a prerequisite for ensuring the safety and efficiency of designing deep foundations. (Ren & Sun, 2023) Traditionally, geotechnical engineers rely on various methods in order to estimate the value of ultimate bearing capacity. Nonetheless, most of the methods used for estimating the ultimate bearing capacity are either analytical or empirical. The analytical or empirical approaches are cost-effective, but they are also site-specific and tend to oversimplify the scenario. (Randolph, 2003) To validate the analytical results, a static load test (SLT) is required to be carried out. The static load test (SLT) is a reliable method for accurately depicting soil-structure interactions. However, it is laden with several limitations, including its high cost and time-consuming aspect. (Momeni et al., 2013)

Given the increasing complexity of foundation design, more sophisticated ultimate bearing capacity (UBC) estimation approaches are critical. Numerical approaches using finite element software like PLAXIS 3D offer a potent option for investigating complicated soil-pile interactions. This method reduces time and cost compared to field tests and allows for numerous analyses under various soil conditions and pile types. This explains the recent popularity of numerical simulations in geotechnical and structural analysis. (Ozturk et al., 2023) & (Krasiński & Wiszniewski, 2017)

PLAXIS 3D is one of several finite element software options for geotechnical analysis. This thesis examines how well PLAXIS 3D simulations estimate the ultimate bearing capacity of piles under static loading conditions.

This research aims to evaluate the accuracy of PLAXIS 3D in predicting the load-settlement responses of soil-pile interaction behavior during static pile load tests, with a focus on estimating their ultimate bearing capacity. PLAXIS 3D models provide a more detailed representation of soil behavior compared to conventional analytical methods, potentially resulting in more precise predictions of ultimate bearing capacity. The goal is to scrutinize the current limitations of ultimate bearing capacity estimation methods and to contrast the results obtained from PLAXIS 3D with those from traditional methods. Ultimately, the objective is to determine whether PLAXIS 3D offers a more effective and reliable approach to assessing the ultimate bearing capacity.

1.2 Research Objectives and Overview

1.2.1 General Objective:

This study investigates the effectiveness of PLAXIS 3D, a finite element program, in estimating the ultimate bearing capacity of piles under static loading conditions.

The main goal is to assess the accuracy of PLAXIS 3D simulation in estimating the ultimate bearing capacity by comparing the observed estimation to well-established analytical methods that use empirical equations and field tests that use the static load test.

1.2.2 Specific objectives

- To determine the input parameters required for numerical simulation using PLAXIS 3D.
- To validate the finite element method model developed in PLAXIS 3D by comparing the obtained ultimate bearing capacity from both analytical and field test approaches.

1.2.3 Research Overview

The geotechnical engineering professionals traditionally either opt for analytical methods or for field pile load tests for ultimate bearing capacity estimation. Analytical methods are the one with cost efficiency but it lacks the accuracy due to the over-simplifications in soil behavior modelling. On the other hand, the field static pile load test offers a more comprehensive assessment by directly observing pile behavior under static loading, but is more expensive and time consuming.

Modern foundation designs have become increasingly complex. sophisticated methods are required for accurately estimating the ultimate bearing capacity. Finite element software provides a reliable alternative for computational modeling. Finite element software enables the simulation of intricate soil-pile interactions, resulting in a far more precise representation of the behavior when compared to simplified analytical models.

This study examines the efficiency of PLAXIS 3D, a regularly utilized finite element program, in accurately determining the maximum load-bearing capacity of piles under static loading conditions.

1.3 Organization of the thesis

This section should have a brief description of the thesis outline of thesis. It should contain the chapter no. with title and brief descriptions of the content of each chapter. An example guide is provided below.

Chapter 1: Introduction and Objective. This chapter provides the background and motivations of the research. The overall objectives and expected outcomes are also described in this chapter.

Chapter 2: Literature Review. This chapter reviews the related works in the geotechnical field with a special focus on the ultimate bearing capacity of soil for pile.

Chapter 3: Methodology. This chapter describes the methodology adopted to carry out the research.

Chapter 4: Results and Discussion. This chapter describes the results of numerical simulation of static load test

Chapter 5: Conclusions and Future Work. This chapter summarizes the conclusions and major contributions of this study, and provides recommendations for future studies.

CHAPTER 2

Literature Review

2.1 Introduction

The estimation of ultimate bearing capacity of piles in an accurate manner is critical to assure the safety and to enhance the efficiency of deep foundation design. To estimate the ultimate bearing capacity, the geotechnical engineering professionals have to most often remain reliant either on the analytical or empirical method that utilizes the several empirical formulae or on the field tests such as static load test. Nevertheless, the both methods have some constraints. The analytical methods usually oversimplify the intricate soil behavior, while the field methods are impeded by the constraints like being expensive and being time-consuming, along with other type of limitations.

In contrast, PLAXIS 3D is a finite element software which is a potential candidate to provide the solution. As the software allows the professionals to simulate the intricate soil-pile interaction behavior, which provides a better, cost-effective and make the process more streamlined and make reliable estimation of the ultimate bearing capacity. The current literature review helps to determine how effective PLAXIS 3D simulations can be in estimating the ultimate bearing capacity, as compared to analytical methods and field tests.

2.2 Key Findings

2.2.1 Effectiveness of PLAXIS 3D Simulations

Several studies have evaluated the effectiveness of estimating ultimate bearing capacity by using PLAXIS 3D. In light of that study, we briefly discuss the effectiveness of PLAXIS 3D simulations:

- **Comparison with Analytical or Empirical Methods:** The research published by (Momeni et al., 2013) have compared the results obtained from the simulations by using finite element software named PLAXIS 3D with the analytical method, empirical method, PDA (Pile Driven Analyzer) & SLT (Static Load Test) in granular material. The study result illustrates that finite element analysis (a computer simulation) specifically PLAXIS 3D yielded results that agree with other methods for measuring pile capacity like empirical or semi-empirical method, PDA and SLT.

- **Validation with Field Tests:** Studies carried out by (Zuievska et al., 2014) have compared PLAXIS 3D simulations with results from static pile load tests. The reported results demonstrate that, on average, the correlation coefficient was 0.98, and the average difference between maximum values of settlement is 5%, indicating a high reliability of the data obtained in the simulation, representing PLAXIS 3D's ability to capture pile-soil interaction in a consistent manner.

These studies suggest that PLAXIS 3D can be a valuable tool for ultimate bearing capacity estimation, that offers advantages over simplified analytical methods and moreover it is a cost-effective and time-saving option considering the static load test done at the field. However, it's important to consider limitations:

- **Input Parameter Sensitivity:** The accuracy of PLAXIS 3D simulation is heavily reliant on the accurate input parameters for soil properties and pile behavior (Bowers et al., 2019). It is decisive to characterize of soil and pile materials carefully.

- **Model Calibration:** The growing popularity of three-dimensional finite element methods (3D FEM) in the design process reflects their capacity to offer a profound understanding of soil behavior and soil-structure interactions in comparison to conventional methods. However, in order for these models to be fully reliable and to provide an almost accurate representation, it is crucial to have thorough calibration and validation. The study conducted by Soon et al. (2018) showcases the application of 3D Finite Element Method (FEM) in verifying the behavior of single piles subjected to lateral loads. This is achieved by a back-analysis technique, which may be used for design purposes, post-construction assessment, and failure investigation.

- **Impacts of Soil Nonlinearity and Layering:** Analyzing the non-linear behavior of soil is crucial for comprehending and predicting soil responses under various loading conditions. Multiple studies have emphasized the significance of considering soil nonlinearity and stratification in PLAXIS 3D models. PLAXIS 3D effectively simulates large soil deformations and variations in normal and shear stresses, particularly when compared to other methods like the 3D Particle method (Park et al., 2016). It accurately models the response of single piles subjected to lateral soil movements, showing good agreement with laboratory results. The software's embedded pile feature and the Mohr-Coulomb elastic-plastic constitutive model are effective in representing pile-soil interactions (Al-Abboodi et al., 2015).

2.2.2 Comparison with Empirical and Analytical Methods

Several factors influence the choice between PLAXIS 3D simulations and analytical methods:

- **Project Complexity:** For simple pile designs embedded deep into homogeneous soil, analytical empirical methods may be suitable. However, for complex projects involving layered soils or intricate loading conditions, the ability of PLAXIS 3D to model these complexities can be advantageous. The finite element model can capture the intricate soil-structure behavior and reaction more accurately.
- **Resource Availability:** While analytical methods have the benefits of providing faster and less expensive results, PLAXIS 3D simulations require specialized software and expertise, which can impact project timelines and costs.
- **Accuracy and Reliability:** Traditional pile design approaches frequently rely on broad assumptions about soil behavior. These assumptions may not always be valid depending on specific site conditions, resulting in erroneous pile capacity estimations. PLAXIS 3D, on the other hand, adopts a rigorous finite element modeling approach that can effectively evaluate the distinct attributes of a particular soil profile. This enables more accurate and site-specific estimations of pile behavior, in contrast with conventional techniques.

The accuracy of PLAXIS 3D is further supported by studies like those conducted by Majeed and Haider (2018). The studies demonstrated that PLAXIS 3D predictions closely corresponded to piles' accurate behavior. Accuracy is of utmost importance in high-stakes projects where safety and performance are critical.

- **Scalability and Adaptability:** PLAXIS 3D is flexible and may be customized to fit projects of different scales, ranging from small residence foundations to huge infrastructure projects. Due to its scalability and versatility, this tool is highly adaptable and can be used to address various geotechnical difficulties. It offers customized solutions that are adapted to meet the individual requirements of each project.

2.2.3 Comparison with Field Tests

Field tests, particularly static load tests, remain the benchmark for ultimate bearing capacity estimation. They provide direct in-situ measurements of pile behavior under static load, presenting the most reliable assessment of ultimate bearing capacity. However, field tests can be expensive and time-consuming. Additionally, Constraints such as limited accessibility or the existence of subterranean utilities may render on-site testing unfeasible.

Analytical approaches are frequently viewed as an economical means to estimate the ultimate load-bearing capacity during the preliminary design phase. Nevertheless, both analytical and empirical methods require validation through static load test (SLT) observations since they have a propensity to oversimplify intricate soil-structure interactions, resulting in unreliable estimations.

PLAXIS 3D offers a comprehensive and highly reliable approach to capturing the complicated behavior of soil-structure interactions, particularly during the preliminary design phase. PLAXIS 3D is a useful alternative to typical analytical and empirical approaches, effectively overcoming their limits.

Utilizing numerical simulations with PLAXIS 3D to estimate ultimate bearing capacity has the potential to serve as a replacement or addition to traditional methods. The estimates obtained from PLAXIS 3D can be validated by comparing them with static load test (SLT) results and, if necessary, by comparing them with analytical or empirical estimates along with static load test observations (SLT). This approach enhances comprehension of the interactions among soil and structure.

Therefore, employing a comprehensive approach that incorporates both field tests and PLAXIS 3D simulations is the most efficient method for accurately determining the ultimate bearing capacity. Field tests are persisted to be necessary to verify the accuracy of PLAXIS 3D results and the estimations from the analytical or empirical method. Nevertheless, PLAXIS 3D is a helpful tool for use in the preliminary design phases or in complicated situations when conducting on-site experiments is not feasible. By employing this dual technique, the utilization of field testing may be optimized, resulting in more economical and precise estimations of the ultimate bearing capacity for the design of deep foundations.

2.3 Summary

PLAXIS 3D simulations present a powerful tool for estimating UBC of piles subjected to static incremental loading. Compared to analytical methods, PLAXIS 3D can account for complex soil behavior, potentially leading to more accurate results. However, careful selection of input parameters and model calibration are crucial for reliable predictions. While field tests remain the most realistic approach, PLAXIS 3D offers a valuable alternative for many projects due to its efficiency and ability to model complex scenarios. This review establishes the potential of PLAXIS 3D simulations and highlights areas for further research in your specific project.

2.4 Future Research Directions

Although the existing literature shows the promise of PLAXIS 3D, there are some areas that need additional research in order to fully utilize its capabilities:

- **Improved Soil Models:** Future research should focus on developing more accurate soil models that can accurately show how different types of soil behave under different loading situations. Utilizing advanced soil models will improve PLAXIS 3D simulations' dependability and produce a more accurate final bearing capacity estimation.
- **Machine learning incorporation:** The integration of machine learning algorithms with PLAXIS 3D has the potential to improve its predictive skills. Machine learning can enhance the identification of patterns and correlations in

extensive datasets, resulting in enhanced precision of input parameters and superior simulation outcomes.

- **Real-time Data Integration:** Real-time data integration entails devising techniques to incorporate data from field sensors into PLAXIS 3D simulations in a timely manner. This integration would allow for the continual updating and improvement of the models. This real-time methodology would facilitate a more dynamic and responsive geotechnical study.
- **Cross-disciplinary Collaboration:** Collaboration between geotechnical engineers, software developers, and data scientists can result in the development of novel solutions and enhancements in PLAXIS 3D simulations. Interdisciplinary research has the potential to propel progress in modeling methodologies and software capabilities.
- **Case Studies on Diverse Soil Conditions:** Conducting additional case studies in various soil conditions and geographical regions would enhance the validation and refinement of PLAXIS 3D's applicability. These studies can offer significant perspectives on the software's performance in various environments and project categories.
- **Education and Training Initiatives:** Creating extensive educational programs and training modules focused on PLAXIS 3D can provide engineers with the essential expertise to proficiently utilize the software. These programs should encompass advanced modeling approaches, calibration methodologies, and optimal strategies for incorporating field data.

2.5 Conclusion

The literature review underscores the significant potential of PLAXIS 3D in estimating the ultimate bearing capacity of piles. Its ability to model complex soil-pile interactions, coupled with advancements in computational power, makes it a valuable tool in geotechnical engineering. While field tests and analytical methods remain crucial, the integration of PLAXIS 3D into geotechnical practice offers a more comprehensive, accurate, and cost-effective approach to foundation design. Continued research and development in this field will further enhance the reliability and applicability of PLAXIS 3D, leading to safer and more efficient deep foundation systems.

CHAPTER 3
Methodology

3.1 Introduction

The objective of this study is to conduct a thorough evaluation of different approaches for estimating the ultimate bearing capacity (UBC) of piles. Piles are crucial deep foundation elements used in civil engineering to transfer loads from superstructures to deeper, more stable soil layers or rock formations. Accurate estimation of the ultimate bearing capacity (UBC) of piles is required for ensuring the safety, stability, and longevity of structures such as buildings, bridges, and towers. The ultimate bearing capacity alludes to the maximum load that a pile can withstand without failing or incurring excessive settling. This crucial factor influences the design and construction methods for foundations, directly affecting their structural integrity and safety. In the past, people widely recognized static load tests (SLTs) as the most reliable method for evaluating the ultimate bearing capacity (UBC) of piles.

Static Load Tests (SLTs) provide instantaneous, reliable, and accurate measurements of pile behavior under controlled loading conditions. During a static load test (SLT), the test pile is subjected to an increasing, gradual load in stages. Each increase in load is sustained for a sufficient amount of time to observe how the pile reacts. Sophisticated instruments are used to carefully measure the settlement of the pile tip after each load increment. The careful compilation of data leads to a comprehensive and detailed load-settlement curve, which is used as the foundation for estimating the UBC. To further illustrate this process, a flowchart can be included to depict the sequence of steps involved in conducting SLTs and collecting the load vs. settlement (Q-z) curve. This visual aid will help clarify the methodical procedure and highlight the critical stages of data collection and analysis.

Process diagram:

STATIC LOAD TEST

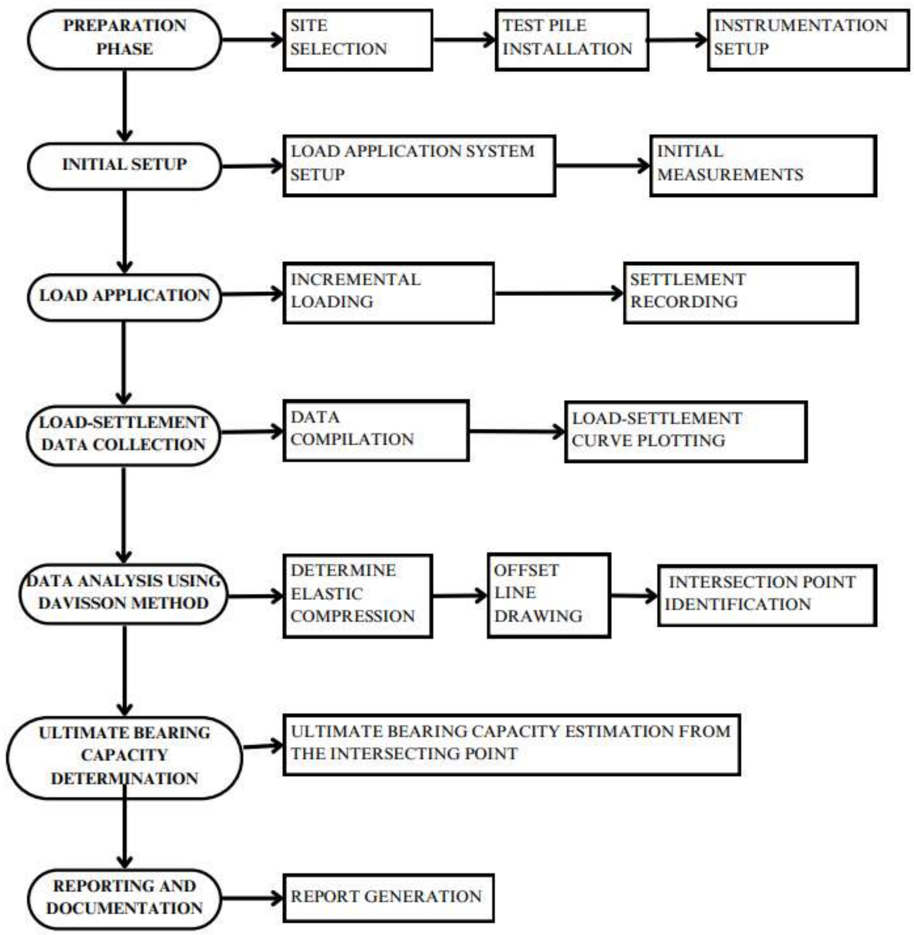

Figure 3-1-1. Process diagram and flowchart for Static Load Test

The load-settlement curve that results from a static load test (SLT) is investigated using several interpretive techniques to ascertain the ultimate bearing capacity. The Davisson method (1972) is an established method that specifies the ultimate bearing capacity as the load at which the settlement intercepts the elastic compression of the pile, together with a factor associated with the pile diameter. The Davisson method is valued for its simplicity and practicality, offering a distinct standard for determining the ultimate bearing capacity (UBC) based on the load-settlement reaction.

Despite their high accuracy and reliability, static load tests (SLT) are often resource-intensive, time-consuming, and financially expensive. The extensive preparation, specialized equipment, and labor required to conduct static load tests (SLTs) make them expensive and resource-intensive. Additionally, the need for a controlled environment and precise instrumentation further adds to the logistical challenges associated with static load testing (SLT). Considering these limitations, various methods for estimating the ultimate bearing capacity (UBC) of piles have been devised and are being gradually implemented in practice. The approaches can be classified into three main categories: empirical, analytical, and in-situ testing procedures. Each strategy has unique benefits and constraints, making them appropriate for varying project specifications and site circumstances.

This study aims to investigate the effectiveness of different UBC estimation methods and their suitability for practical engineering applications. The focus is on three primary approaches:

1. **Analytical Methods:** Analytical methods utilize theoretical frameworks and recognized mathematical equations to estimate the ultimate bearing capacity (UBC) of piles based on soil parameters (such as cohesion, friction angle, and effective stress) and pile design (including diameter and length). These approaches offer a well-defined and systematic framework for estimating ultimate bearing capacity (UBC), which can be utilized for various soil conditions and types of piles.

   These methods frequently incorporate assumptions regarding soil behavior and soil-pile interactions. This enables engineers to formulate mathematical equations that can estimate the capacity of piles to sustain loads. The assessments take into account factors such as soil stratigraphy, pile material, and the process of pile installation to improve the accuracy of the predictions. Analytical methods can be particularly useful in the preliminary design stages, offering quick estimates that inform decision-making processes.

   Although analytical approaches provide a solid theoretical foundation for estimating ultimate bearing capacity (UBC), their accuracy is highly dependent on the accurate characterization of soil parameters. To obtain accurate values for soil parameters such as cohesion, friction angle, and effective stress, thorough site investigations and laboratory testing are necessary. Obtaining accurate and comprehensive soil data can be difficult in practice, mostly because of the inherent variations in soil characteristics and the limitations of testing techniques. If the soil parameters are not well-characterized or if there is high variation within the soil profile, it may undermine the credibility of analytical methods.

2. **Empirical Methods:** Empirical methods rely on the correlations between readily available in-situ test data (such as the Standard Penetration Test, SPT) and field observations. These approaches provide an effective and affordable way to estimate the ultimate bearing capacity (UBC) of piles, particularly in situations where doing extensive soil investigations is not practical or feasible.

   Empirical methods usually rely on extensive datasets containing in-situ test data (such as the Standard Penetration Test, SPT) and field observations. Empirical formulas relate the soil's resistance to penetration, measured by the Standard Penetration Test (SPT), to the ultimate bearing capacity (UBC) of piles. Moreover, Cone Penetration Test (CPT) data can estimate empirical correlations for the ultimate bearing capacity (UBC).

   Empirical methods have the significant benefit of being simple and straightforward to put into practice. Compared to more intricate analytical or in-situ testing methods, these approaches require minimal site preparation and can be executed relatively expeditiously. In addition, empirical approaches are economical, making them an appealing option for preliminary design and feasibility studies.

   However, the implicit assumptions and simplifications in the correlations may restrict the accuracy of empirical approaches. Typically, we establish these correlations using precise soil conditions and pile types, which may limit their application to similar situations. Empirical approaches may produce less accurate results when employed on soil conditions that differ significantly from those used to establish the correlations. Hence, it is essential to exercise prudence and take into account site-specific variables while employing empirical methods for ultimate bearing capacity estimation.

3. **PLAXIS 3D Simulations:** Simulations with PLAXIS 3D PLAXIS 3D is a sophisticated program that uses finite element analysis to accurately represent the interaction between soil and structure. It can simulate a wide range of loading conditions and soil profiles. This simulation tool has the ability to optimize the design process by offering an economical and time-efficient procedure for ultimate bearing capacity (UBC) estimation.

   PLAXIS 3D employs advanced computational techniques to model the behavior of both the pile and the surrounding soil. The software has the ability to replicate intricate interactions between the pile and the soil, taking into consideration aspects like soil heterogeneity, non-linear material behavior, and different loading conditions. Engineers can gain a comprehensive understanding of a pile's load-bearing characteristics and estimate its ultimate bearing capacity (UBC) by inputting specific soil data as well as pile shape.

   PLAXIS 3D simulations offer a significant advantage in their capacity to accurately represent a diverse array of situations and loading conditions. Engineers can use this adaptability to explore various design alternatives and improve the pile design to suit specific site conditions. Furthermore, PLAXIS 3D has the capability to provide valuable insights into the behavior of piles when subjected to extreme loading conditions, such as seismic occurrences or heavy traffic loads, which can be difficult to replicate in physical tests.

   To enhance the reliability of PLAXIS 3D simulations, engineers must ensure that the input data, such as soil properties and pile dimensions, are as accurate and detailed as possible. Additionally, the calibration process should involve comparing simulation results with field observations or experimental data to refine the model parameters. This iterative process helps in achieving more precise and realistic predictions of the pile’s ultimate bearing capacity.

   Overall, while PLAXIS 3D offers a powerful tool for UBC estimation, its effectiveness depends on the meticulous preparation of input data and thorough calibration efforts. By addressing these factors, engineers can leverage PLAXIS 3D to develop safer and more efficient pile designs, ultimately contributing to the stability and longevity of civil engineering structures.

The method's goal is to determine how well and accurately each method can reproduce real-life pile behavior using a framework for comparing a large amount of recorded static load test (SLT) data. The systematic and gradual incremental application of the load on the test pile allows for a comprehensive assessment of how the pile reacts to increasing stress. This serves as a useful reference point for evaluating the accuracy of estimations made by different ultimate bearing capacity (UBC) estimation methods.

If the predicted ultimate bearing capacity (UBC) values match the measured data from the static load test (SLT) and the load-settlement curves have a similar overall shape, then the suggested method correctly shows the important parts of how piles behave under static loading. However, this section will thoroughly explore the inherent limitations of all the assessed approaches.

3.2 Site Location

The Static Load Test (SLT) and Geotechnical Investigation for this study were carried out at the site of proposed Samabaya Bhaban Plot No. F/10, A & B, Sher-e-Bangla nagar, Dhaka.

Geotechnical investigations at the site included the data collected from Standard Penetration Tests (SPT) and laboratory testing of soil samples to determine properties such as cohesion and friction angle. These investigations provided the data of soil profiles and critical parameters necessary for accurate ultimate bearing capacity estimation using analytical, empirical, and simulation methods.

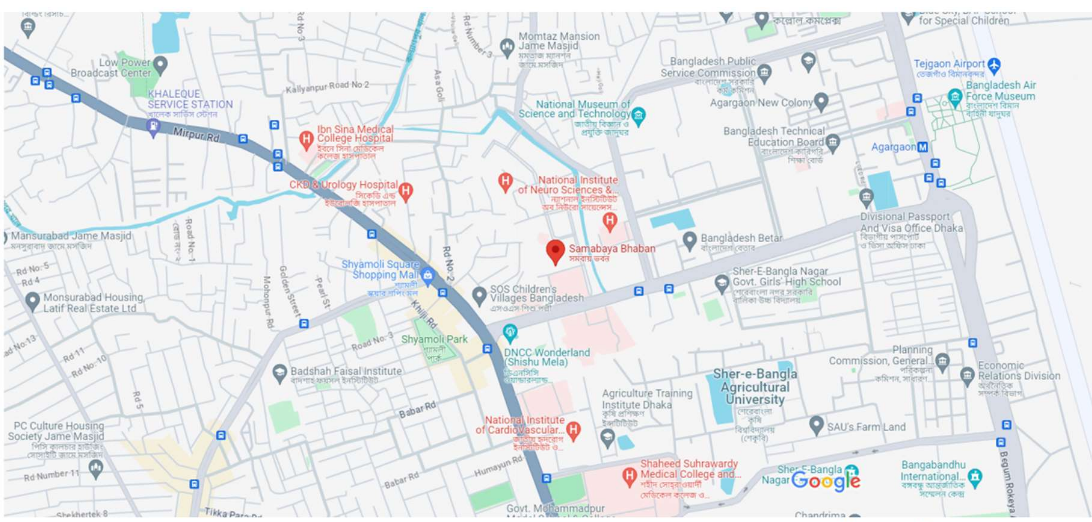

Figure 3-2-1. Site location

3.3 Methodology Overview

3.3.1 Data Collection

3.3.1.1 *Detailed Test Setup:*

To begin evaluating the estimation methods for ultimate bearing capacity (UBC), it is necessary to obtain a reliable dataset from extensively reported static load tests (SLTs). The data collection procedure is crucial as it serves as the basis for evaluating the accuracy provided by various analytical, empirical, and numerical methods. The primary focus is on obtaining comprehensive information about the test setup, observed pile behavior, and relevant soil data. The following components are described in detail:

The elaborate test setup entails gathering extensive data on the pile's geometry, the test location's conditions, and the specific loading technique used throughout the static load tests (SLTs). Critical elements include the following:

1. **Pile Geometry:** Accurate data regarding the dimensions and material properties of the piles is of paramount significance. This includes the dimensions of the pile, including its diameter, length, and the specific material utilized. In this investigation, the test piles have been defined by a diameter of 22 inches (558.8 mm) and a length of 43 feet (13.00 m). The piles are cast-in-situ reinforced cement concrete (RCC) piles.

2. **Test Site Conditions**: During the SLTs (Site Location Tests), it is necessary to keep an accurate record of the site conditions. This includes information about the specific location, the ground surface, and any relevant environmental elements that may affect the test's results.

3. **Loading Protocol:** The selection of the loading protocol for imparting the load to the piles during the SLTs is of critical significance. This investigation divided the total load into eight increments. We maintained each increment until the rate of settlement was smaller than 0.25 mm/hr or until 2 hours had passed, whichever came first. Once the maximum load was attained, it was sustained for a minimum of twelve hours before being gradually reduced in four increments

The accurate documentation of the pile installation procedure is also essential. This includes documentation of the driving or boring technique employed, the equipment used, and any challenges encountered during the installation process. This information aids in comprehending the initial conditions and the interplay between the pile and the surrounding soil.

3.3.1.2 *Measured Pile Behavior:*

The primary focus here is on acquiring and analyzing the total load-settlement curves documented during the static load tests (SLTs). These curves provide significant insight into the pile's behavior as loads increase, enabling the identification of critical factors such as the ultimate load and the corresponding settlement. The information from the SLTs (static load tests) is very important for making sure that the methods used to estimate ultimate bearing capacity (UBC) are correct.

Load-Settlement Data

Below is the detailed data obtained from the SLTs, which includes the applied maximum load, settlement at maximum load, net settlement, and ultimate load using the Davisson Method.

Table 3-1. Summary of Static Load Test (SLT) Results and Ultimate Load Calculation Using Davisson Method

S1. No.	Pile No.	Applied Maximum Load (kip)	Settlement at Maximum Load (mm)	Net Settlement (mm)	Ultimate Load Using Davisson Method (ton)
1	TP-4	187.5	14.4	8.5	186.0
2	TP-1	204.7	25	22	195.0
3	TP-2	196.0	24.5	20.5	172.0

Detailed Analysis

1. **Pile TP-4:**
    - **Applied Maximum Load:** 187.5 kips
    - **Settlement at Maximum Load:** 14.4 mm
    - **Net Settlement:** 8.5 mm
    - **Ultimate Load (Davisson Method):** 186.0 tons
2. **Pile TP-1:**
    - **Applied Maximum Load:** 204.7 kips
    - **Settlement at Maximum Load:** 25 mm
    - **Net Settlement:** 22 mm
    - **Ultimate Load (Davisson Method):** 195.0 tons
3. **Pile TP-2:**
    - **Applied Maximum Load:** 196.0 kips
    - **Settlement at Maximum Load:** 24.5 mm
    - **Net Settlement:** 20.5 mm
    - **Ultimate Load (Davisson Method):** 172.0 tons

These measurements are essential to perceive how the pile behaves when subjected to static loading conditions. The load-settlement data is used to create load-settlement curves, which are vital for comparing the estimated ultimate bearing capacity (UBC) acquired from different analytical, empirical, and numerical methods with the actual measurements obtained from SLTs.

3.3.1.3 *Relevant Soil Data:*

Relevant soil data is essential for a thorough investigation. This encompasses in-situ test outcomes such as Standard Penetration Test (SPT) N-values and laboratory analysis of soil properties such as unit weight and shear strength parameters. The SPT values ranged between 2 and 8 in the clayey soil layer, which stretched up to a depth of 20–25 feet. In the sand layer, the SPT values ranged from 8 to 50 at depths of 25–45 feet. At depths less than 45 feet, the Standard Penetration Test (SPT) results exceeded 50, indicating the existence of highly compacted soil conditions. Comprehensive soil profiles, which account for differences in soil type and geology, are necessary for accurate estimation of ultimate bearing capacity (UBC) by analytical, empirical, and numerical methods.

Data Collection Process

1. **Site Investigation:** Conducting a thorough site investigation is critical to gather comprehensive soil data necessary for the evaluation of ultimate bearing capacity (UBC) estimation methods. The site investigation includes the following steps:

    - **Drilling Boreholes:** Ten boreholes were drilled across the test site to explore the subsurface conditions. The boreholes provided detailed information about the soil strata, which is essential for understanding the soil profile and its variability.

    - **In-Situ Tests:** Various in-situ tests were performed to obtain soil properties directly from the field. The Standard Penetration Test (SPT) and Cone Penetration Test (CPT) are the primary in-situ tests conducted. The SPT values ranged between 2 and 8 in the clayey soil layer up to a depth of 20–25 feet, from 8 to 50 in the sand layer at depths of 25–45 feet, and exceeded 50 below 45 feet, indicating highly compacted soil conditions.

    - **Laboratory Testing:** Soil samples collected from the boreholes were subjected to laboratory tests to determine soil properties such as unit weight, cohesion, friction angle, and shear strength parameters. These properties are crucial for analytical and empirical methods of UBC estimation.

- **Borehole Logs and Stratigraphic Profiles:** Detailed borehole logs were created to document the findings from the drilling process. Stratigraphic profiles were developed to visualize the subsurface conditions, showing the distribution of different soil layers and their properties.

2. **Pile Installation Records:** Accurate documentation of the pile boring procedure is essential for understanding the initial state of the pile and its interaction with the surrounding soil. During the pile installation process, we recorded the following data:

   - **Installation Method:** To install the piles, we used either the driven or bored method. In this specific case, we employed cast-in-situ reinforced cement concrete (RCC) piles.

   - **Installation Depth:** To achieve the desired soil stratum that can support the load, we installed the piles at a depth of 43 feet (13.00 m).

   - **Encountered Issues:** Any difficulties or problems experienced throughout the installation procedure were recorded. This encompasses challenges such as driver noncompliance or obstacles encountered during excavation.

3. **SLT Setup and Execution:** Proper setup and execution of the Static Load Test (SLT) are critical to obtaining accurate data on the pile's behavior when it is subjected to loading. The process consists of several detailed steps designed to ensure the integrity and reliability of the test results. The following sections outline these steps in comprehensive detail.

   - **Instrumentation:** The test piles were equipped with load cells, displacement transducers, and other instrumentation that accurately measured the applied load and resulting settlement.

   - **Loading Protocol:** The total load was divided into eight equal halves. The increment was sustained until the settlement rate dropped below 0.25 mm/h or until 2 hours elapsed, whichever occurred first. After reaching the maximum load, it was maintained for at least twelve hours before being gradually decreased in four stages.

- **Load-Settlement Curve:** The load and settlement were recorded at each stage of loading, resulting in a complete load-settlement curve. This curve provides critical insights into the pile's behavior under static loading conditions.

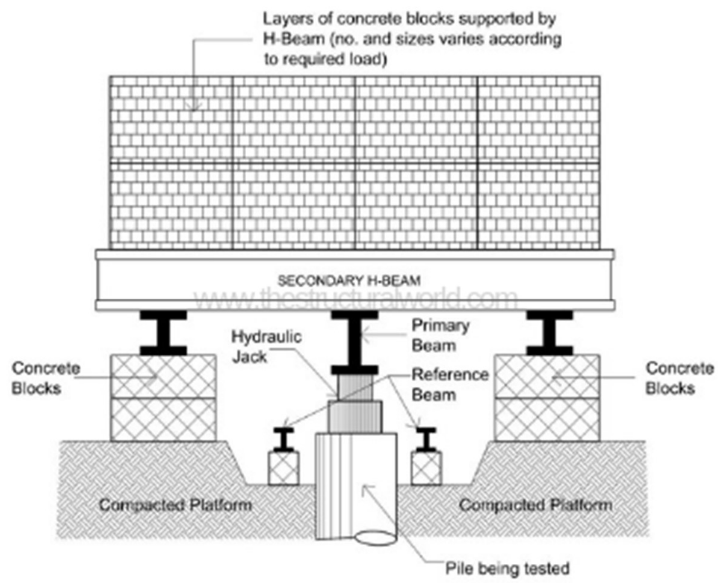

Figure 3-3-1. Scheme of Static Pile Load Test Set-up

3.3.2 Evaluation of Analytical Methods

Analytical methods developed by Vesic (Vesic, A. S. 1977), Meyerhof (Meyerhof, G. G. 1976), and O'Neill and Reese (O'Neill, M. W., & Reese, L. C. 1999) are critical in predicting the axial bearing capacity of piles. Each method offers a distinct approach based on theoretical principles and empirical correlations derived from field observations and laboratory data.

Theoretical Framework

Analytical methods for predicting the axial ultimate bearing capacity of piles are grounded in fundamental principles of soil mechanics and foundation engineering. These methods utilize theoretical frameworks to establish relationships between soil properties, pile geometry, and the ultimate load-bearing capacity of the pile. This section outlines the key theoretical aspects underlying analytical methods developed by prominent researchers such as Vesic, Meyerhof, and O'Neill and Reese.

Fundamental Principles

- **Effective Stress Concept:** Effective stress is a fundamental principle in soil mechanics. It states that the ability of soil to sustain loads is determined by the effective stress applied to the soil, which determines its ability to sustain loads. This effective stress is calculated by subtracting the pore water pressure from the total stress. Analytical approaches utilize the principles of effective stress to evaluate the contributions of both end-bearing and skin friction to the axial capacity of piles.

- **Shear Strength Parameters:** Soil shear strength parameters, namely cohesion (c) and angle of internal friction (φ), play pivotal roles in assessing the soil's resistance to both pile penetration and lateral deformation. Cohesion represents the soil's inherent strength due to particle adhesion, while the friction angle indicates the resistance to sliding between soil particles. These parameters are fundamental for evaluating the stability and load-bearing capacity of deep foundation elements like piles.
  To determine these crucial parameters, engineers often rely on empirical correlations with Standard Penetration Test (SPT) N values. The SPT involves driving a split-barrel sampler into the soil and recording the number of blows required for each increment of penetration. These N values provide a standardized measure of soil resistance at different depths.
  Given concerns about the reliability of laboratory testing for cohesion, which can be influenced by factors like sample disturbance or variability, empirical correlations offer a practical alternative. These correlations establish relationships between SPT N values and shear strength parameters, enabling engineers to estimate cohesion and friction angle directly from in-situ test data. This approach enhances the efficiency and reliability of geotechnical assessments, supporting informed decision-making in foundation design and construction.

- **Load Transfer Mechanisms:** The load transfer mechanisms in piles comprise end-bearing resistance at the pile tip and skin friction along the pile shaft. Analytical methods measure and calculate these mechanisms utilizing empirical factors and coefficients obtained from observations in the field and theoretical considerations.

O'Neill and Reese Method

The load-settlement behavior of a drilled shaft under short-term compression loading can be calculated using normalized relation curves proposed by O'Neill and Reese (1999). The method integrates the analysis of load transfer mechanisms in drilled shafts, combining end-bearing resistance and side friction to predict the axial capacity. These normalized relation curves are derived from extensive field load tests and provide a practical means to estimate the load-settlement response of drilled shafts.

- **For Total Stress Analysis Method (For Cohesive Type of Soil)**
- **Adhesion**
  - — $\alpha_u = 0.55;\ \frac{S_u}{P_a} \leq 1.5$
  - — $\alpha_u = 0.55 - 0.1 \times \left(\frac{S_u}{P_a} - 1.5\right);\ 1.5 \leq \frac{S_u}{P_a} \leq 2.5$
  - — $f_s = \alpha_u \times S_u \leq 380\ kPa$
- **End Bearing**
  - — $f_b = N_c \times (S_u)_b;\ N_c = 6 \times (1 + 2\frac{z}{D});\ N_c \leq 9;\ f_b \leq 4\ MPa$
  - — Where, z is the embedded depth of the drilled shaft in the end bearing layer and $(S_u)_b$ is the average undrained shear strength over two diameters below the base.
- **For Effective Stress Analysis Method (For Cohesive Type of Soil)**
- **Skin-Friction**
  - Fine-grained Soils:
    - — $\beta = (1 - \sin(\phi'_{cs})) \times (OCR)^{0.5} \times \tan(\varphi_i)$
  - Coarse-grained Soils:
    - — $\beta = 1.5 - 0.245 \times \sqrt{z};\ N_{60} \geq 15$
    - — $\beta = \frac{N_{60}}{15} \times (1.5 - 0.245 \times \sqrt{z});\ N_{60} < 15$
    - — $f_s = \beta \times \sigma'_z \leq 200\ kPa$
    - — z = depth (m) measured from the ground surface to the middle of the soil layer.

- Gravels and Sandy Gravels
  - $\beta = 1.5 - 0.15 \times z^{0.75}; z = depth\ (m)$
  - $\beta = 0.25$ for $z > 26\ m$
  - $f_s = \beta \times \sigma_z' \leq 200\ kPa$

- **End Bearing:**
  - $f_b = 57.5 \times N_{60}; f_b \leq 2900\ (kPa); \frac{L}{D} \geq 10$
  - $f_b = 5.75 \times L \times N_{60}; f_b \leq 290L\ (kPa); \frac{L}{D} < 10$

The O'Neill and Reese method integrates end-bearing resistance and skin friction components to provide a comprehensive estimation of axial capacity and load-settlement behavior of drilled shafts. This method utilizes normalized load-transfer curves derived from extensive field load tests, offering a practical and reliable approach to predict the performance of drilled shafts under short-term compression loading. The O'Neill and Reese method is particularly applicable to both cohesive and cohesionless soils, where empirical factors such as the adhesion factor (α) for cohesive soils and the lateral earth pressure coefficient (K) for cohesionless soils play significant roles in determining the shaft's ultimate capacity and settlement characteristics.

Calculation Procedures

- **Soil Parameter Determination:** The determination of crucial soil parameters such as undrained shear strength (c_u), angle of internal friction (φ), and effective stress (σ') is vital for accurate foundation design. These parameters are derived using empirical correlations with Standard Penetration Test (SPT) N values obtained from in-situ tests. This approach enhances the reliability of geotechnical assessments by leveraging standardized soil resistance measurements at various depths.

- **Undrained Shear Strength (c_u):**

  - **For cohesive soils:**

    - Sanglerat (1972) was the first to present the q_u-N (SPT) correlation for fine-grained soils, considering the plasticity index of clay soils, and categorizing them into clay and silty clay.

    - $C_u = \frac{25 \times N}{2}$

  - **For silty Clay:**

    - $C_u = \frac{20 \times N}{2}$

- **For Silty loam:**

Table 3-2. Soil parameters needed for the mass wasting module: soil cohesion, angle of internal friction (Lindeburg 2001; Koloski et al. 1989), and porosity (Maidment 1992). Both the maximum and minimum values of the angle of internal friction were used to obtain a normal distribution.

Soil Types	Soil Cohesion (kPa)	Angle of Internal friction (min)	Angle of Internal friction (max)	Porosity
Sand	0	30	35	0.43
Silty Loam	21	30	40	0.48
Loam	23	30	40	0.43
Silty Clay Loam	15	15	30	0.43
Clay	13.5	20	30	0.38
Talus	0	45	45	0.1

- **Angle of Friction (φ)**

  - **For Silty Sands**

Table 3-3. Angle of Friction for Different Soil Types (Shenbagavalli, Ganapathy, Abirami, & Kalaiselvi, 2017)

Soil Type	Angle of Friction
Sand and gravel mixture	33-36
Well graded sand	32-35
Fine to medium sand	29-32
Silt sand	27-32
Silt (non-plastic)	26-30

  - **For cohesionless soils**

    - Early work on estimating qu from N_{60} attempted direct correlations (Meyerhof 1956; Peck et al. 1974). Peck et al.'s (1974) graphical representation was approximated by Wolff (1989) as:

$$\varphi' = 27.1 + (0.3 \times N_{60}) - (0.00054 \times N_{60}^2)$$

    - Gibbs and Holtz (1957) showed that effective overburden pressure (σ′) affects N. Schmertmann (1975) included σ′ in an N−q_u relationship, later approximated by Kulhawy and Mayne (1990):

$$\varphi' = \tan^{-1}[\{\frac{N_{60}}{12.2 + 20.3 \times (\frac{\sigma_0'}{p_a})}\}^{0.34}]$$

Load-Carrying Capacity Calculation: The O'Neill and Reese Method is employed to calculate both skin friction and end-bearing capacity using normalized relation curves. The total ultimate bearing capacity (Q_{ult}) is determined by summing the contributions from skin friction (Q_s) and end-bearing resistance (Q_b). These steps ensure that foundation design accounts for soil-specific characteristics, thereby optimizing structural stability and performance under varying load conditions.

3.3.3 Evaluation of Empirical Methods

The application of empirical correlations, particularly those based upon Standard Penetration Test (SPT) N-values, is one of many methods for estimating the ultimate bearing capacity (UBC) of piles. The empirical methods used in this study are extensively documented, with explicit references to specific methodologies and a comprehensive discussion of their applicability and limitations, ensuring a thorough review.

Empirical Correlations

1. **Meyerhof's Correlation:**

Meyerhof's empirical correlation is a well-acknowledged method for estimating the ultimate bearing capacity (UBC) of piles, especially in granular soils. This method relies on the correlation between the N-values derived from the Standard Penetration Test (SPT) and the ultimate bearing capacity. Substantial field data and research establish the correlation. Meyerhof's technique is an invaluable tool in geotechnical engineering because of its direct and practical approach.

- **Comprehensive Field Data Basis:** Meyerhof's connection is based on a substantial amount of empirical data obtained from extensive field testing and observations. The extensive dataset ensures that the correlation is strong and dependable, giving engineers confidence in its suitability for various soil conditions, particularly granular soils.

- **Straightforward Application:** The application is direct and simple, as it streamlines the estimation of the ultimate bearing capacity (UBC) by utilizing readily available SPT N-values. Its practicality makes it a very efficient option for initial design and evaluation, minimizing the necessity for more complex and time-consuming methods. The simplicity of implementation enables quick decision-making in the field.

- **Consideration of Pile Tip Resistance:** Meyerhof's correlation incorporates the consideration of pile tip resistance, which accounts for the soil's resistance at the tip of the pile. The pile tip resistance plays a crucial role in determining the overall bearing capacity. The procedure computes the tip resistance (q_p) using the formula.

$$Q_b = f_b \times A_b$$

$$f_b = C \times N_{60}$$

$$C = 38 \times \frac{L_s}{D} \leq 380$$

L_S is length of the pile driven in sand

- **Incorporation of Skin Friction:** Meyerhof's method incorporates the consideration of skin friction, which refers to the resistance between the soil and the surface of the pile shaft. We calculate the skin friction (f_{ave}) by averaging the SPT N-value along the entire length of the pile. The equation for calculating the average unit skin resistance is as follows:

$$Q_f = f_s \times Perimeter \times length$$

Displacement Piles: $f_b = 1.9 \times N_{60}; f_s \leq 100kPa$

Non-displacement Piles:

$$f_b = 0.95 \times N_{60}; f_s \leq 50kPa$$

Where $\overline{N}$ is the average SPT N-value. By including both the tip resistance and the skin friction, Meyerhof’s correlation provides a comprehensive assessment of the pile’s bearing capacity.

- **Total Ultimate Bearing Capacity:** The ultimate axial capacity (Q_u) of the pile is estimated by summing the contributions from the pile tip resistance and the skin friction. The formula used is:

$$Q_u = Q_b + Q_f$$

Where:

— A_b is the area of the pile tip.

— Q_u is the ultimate bearing capacity.

— p is the perimeter of the pile.

— L is the length of the pile.

This combined approach ensures that all critical factors influencing the pile capacity are considered, resulting in a more accurate estimation of the ultimate bearing capacity (UBC).

- **Practical Insights and Limitations:** Although Meyerhof's correlation is extremely valuable, it is important to recognize its limitations. The approach possesses a high degree of accuracy when applied to granular soils, but its reliability may be reduced when dealing with cohesive soils or soil profiles that are heterogeneous. Furthermore, the correlation presupposes that the soil qualities and conditions are reasonably homogeneous within the specified depth ranges, which may not always be true in intricate geotechnical situations.

Overall, Meyerhof's correlation provides a balanced approach by integrating empirical data with practical application techniques. Its comprehensive nature, considering both tip resistance and skin friction, makes it a preferred method for estimating the ultimate bearing capacity of piles in many geotechnical projects.

2. **Peck, Hanson, and Thornburn's Recommendations:**

These recommendations offer empirical correlations for determining the load-bearing capacity of piles using data from Standard Penetration Tests (SPT) and Cone Penetration Tests (CPT). The instructions include recommendations to select suitable correlation factors based on soil type and the method used for pile installation. This approach integrates modifications for varying soil conditions and installation methods to enhance the precision of UBC estimations.

Application Procedures

The selection of appropriate correlation factors is a crucial step in accurately estimating the ultimate bearing capacity (UBC) of piles using empirical methods. The chosen factors depend on several key variables:

1. **Selection of Correlation Factors:**

- **Soil Type**:

- **Granular Soils**: These include sandy and gravelly soils. They generally exhibit higher friction angles and lower cohesion, making the pile capacity primarily dependent on the soil's density and frictional properties.

- **Cohesive Soils**: These include clays and silts. They have higher cohesion and lower friction angles, influencing the skin friction and end-bearing capacity of the pile differently compared to granular soils.

- **Mixed Soils**: These soils contain varying proportions of granular and cohesive materials. Estimating UBC in mixed soils requires careful consideration of the relative contributions of cohesion and friction.

- **Pile Installation Method**:

- **Driven Piles**: Installed by hammering or vibrating the pile into the ground. The installation process often densifies the surrounding soil, increasing the pile's load-bearing capacity.

- **Bored Piles**: Installed by drilling a hole and filling it with concrete. This method is less likely to disturb the surrounding soil but requires careful attention to soil-structure interaction during installation.

- **Site-Specific Conditions**:

- **Stratigraphy**: The layering of different soil types at the site. Each layer's properties must be considered in the correlation.

- **Groundwater Level**: The presence of water affects soil strength and the pile's skin friction. Adjustments may be necessary to account for buoyancy effects and potential reduction in effective stress.

To ensure the empirical correlations are tailored to the specific project conditions, adjustments are made to account for variations in soil properties. This customization helps improve the accuracy of the UBC estimation.

2. UBC Estimation:

Once the correlation factors are selected, empirical correlations are applied to the in-situ test data, primarily SPT N-values, to estimate the UBC. The detailed steps include:

- **Collecting and Analyzing SPT N-values**:

  Obtain SPT N-values from borehole logs, which provide a measure of soil resistance to penetration. These values are critical inputs for the empirical correlations.

- **Calculating q_p and f_{ave} using Meyerhof's Equations**:

- **Pile Tip Resistance (q_p)**:

— $q_p = 40 \times \left(\frac{N}{D/L}\right) \leq 400 \times N$

— This equation considers the SPT N-values and the ratio of pile length (L) to diameter (D) to estimate the ultimate stress at the pile tip.

- **Skin Friction (f_{ave})**:

— $f_{ave} = 2 \times \bar{N}$

— This equation uses the average SPT N-value to estimate the unit skin resistance along the pile shaft.

- **Summing the Skin Friction and End-Bearing Capacities to Obtain the Total UBC:**

— The total UBC (Q_u) is the sum of the pile tip resistance (q_p) and the skin friction along the pile shaft *(f_{ave})*.

— $Q_u = (A_p \times q_p) + (p \times L \times f_{ave})$

— *Here, A_p is the cross-sectional area of the pile, and p is the perimeter of the pile.*

The empirical correlations are applied to the in-situ test data (e.g., SPT N-values) to estimate the UBC. The calculated UBC is compared with the measured data from SLTs to evaluate the effectiveness of the empirical methods.

3.3.4 PLAXIS 3D Model Development and Analysis

Developing and analyzing a 3-D model of the pile-soil system employing PLAXIS 3D is a critical step in comprehending and estimating the reaction of piles subjected to axial loads. This rigorous procedure ensures that the model accurately reflects the actual conditions documented during static load tests (SLTs) and site investigations. Through the process of replicating these conditions, we can continue to evaluate and improve our understanding of pile behavior. This section provides an extensive overview of the systematic procedures employed to construct the model, with specific emphasis on pile geometry, soil conditions, loading conditions, and model calibration.

Here is a flowchart illustrating the overall process of PLAXIS 3D model development and analysis.

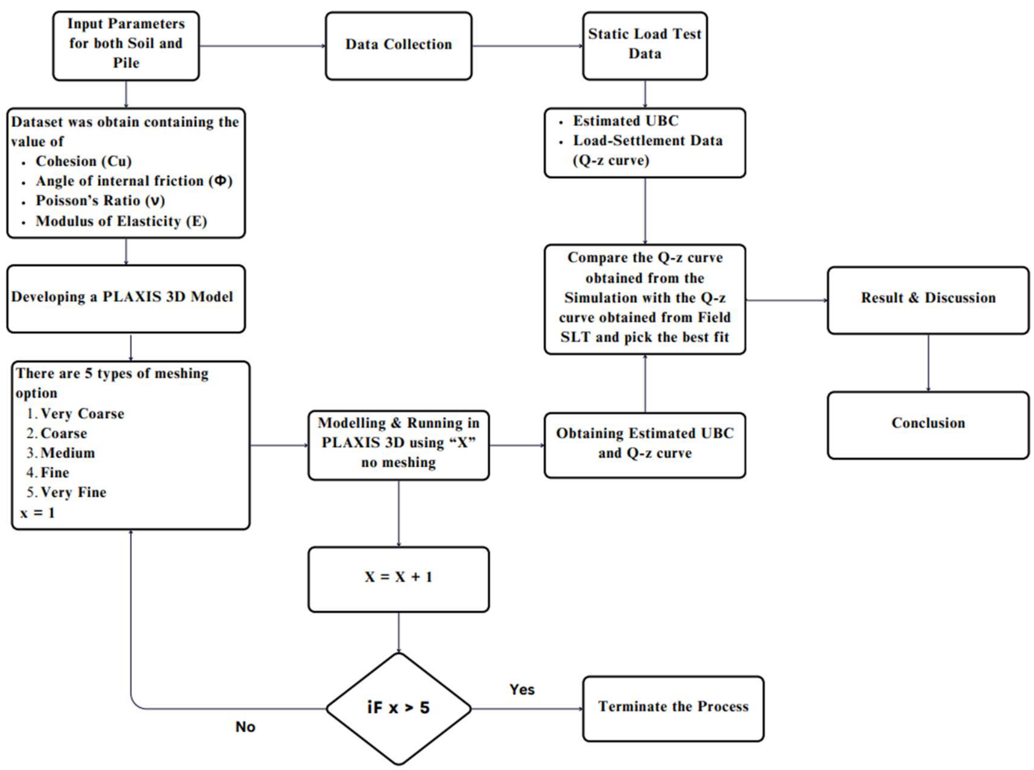

Figure 3-4-1. Framework to carry out PLAXIS 3D simulation modelling.

3.3.4.1 *Pile Geometry*

- ***Dimensions:***

  - **Accuracy in Dimensions:** The precision in modeling the pile's dimensions is paramount to achieving a realistic simulation in PLAXIS 3D. For this study, the pile is specified with a diameter of 22 inches (558.8 mm) and a length of 43 feet (13.00 m). These dimensions are meticulously incorporated into the model to ensure that the geometric representation aligns with the actual physical characteristics of the pile used in the static load tests (SLTs). This accuracy is crucial for simulating the pile's interaction with the surrounding soil, as even slight deviations can significantly impact the load distribution and settlement predictions.

  - **Cross-Sectional Shape:** The cross-sectional shape of the pile plays a vital role in its interaction with the soil. In this study, the cross-section is circular, reflecting the actual pile used in the SLTs. A circular cross-section influences the distribution of stresses and the skin friction mobilized along the pile shaft. Accurately defining this shape in the model ensures that the simulation captures the real-life behavior of the pile under load. If the pile were square or another shape, this would need to be reflected accordingly to ensure the model's accuracy.

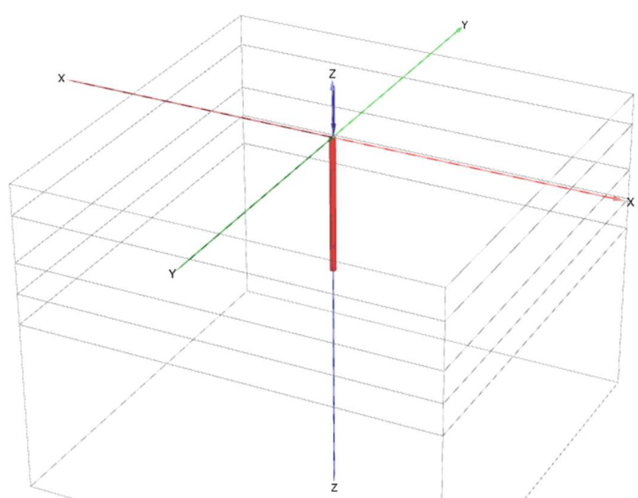

Figure 3-5-1. Geometry of the soil and pile in PLAXIS 3D

- ***Material Properties:***
    - **Young's Modulus (E):** Young's modulus, often denoted as E, is a measure of the stiffness of the pile material, quantifying the relationship between stress and strain in the linear elasticity regime. In this study, the Young's modulus of the concrete pile is specified as 20×10^6 kN/m². This high value reflects the inherent stiffness of concrete, which ensures minimal deformation under applied loads. Accurately specifying Young's modulus in the PLAXIS 3D model is crucial for simulating realistic pile behavior, as it directly influences how the pile deforms under various loading conditions. For instance, a higher E value signifies a stiffer pile that will experience less deformation under the same load compared to a pile with a lower E value.
    - **Poisson's Ratio (ν):** Poisson's ratio, ν, describes the ratio of lateral strain to axial strain in the material when subjected to axial loading. For the concrete pile used in this model, the Poisson's ratio is specified as 0.16. Accurate incorporation of Poisson's ratio in PLAXIS 3D ensures that the model can simulate the lateral contraction of the pile under compression. This lateral deformation affects the interaction between the pile and the surrounding soil, particularly near the pile-soil interface where shear stresses are significant.
    - **Density:** The density of the pile material is a critical factor in determining the self-weight of the pile and its overall stability within the soil matrix. For this study, the density of concrete is specified as 25 kN/m³. Accurate density values are necessary for realistic load distribution within the model, impacting not only the vertical stresses due to the pile's weight but also the overall mass of the system. This is particularly important in static load tests, as the density helps to accurately simulate the self-weight component of the loading and the resulting stress distribution in the soil.

3.3.4.2 *Soil Conditions*

- ***Soil Stratigraphy:*** The soil stratigraphy is a vital component of the model and will be rigorously developed based on borehole logs, SPT data, and soil profiles obtained during the site investigation. In this study, PLAXIS 3D was used to replicate static load tests for Borehole-2 (BH-2) based on the geotechnical investigation report. BH-2 was chosen because it was the nearest to Test Pile-4 (TP-4). The soil properties in BH-2 and TP-4 exhibited a significant level of similarity, which ensured a consistent basis for the modeling process. To particularly represent the underlying conditions, each soil layer will be characterized by its exact thickness, depth, type, and SPT value.

The following stratigraphy is detailed based on the site investigation:

Table 3-4. Soil stratigraphy and Standard Penetration Test (SPT) Values

Depth (m)	Soil Type	SPT Value (N)
0-3.05	Pale yellow ash spotted medium plastic medium clay	2-5
3.05-7.62	Pale yellow ash spotted medium plastic medium clay	5-19
7.62-9.14	Light reddish yellow medium dense silt	19-27
9.14-13.72	Yellow medium dense to dense silty fine sand	27-50
13.72-30.48	Yellow to light reddish yellow and pale brown dense to very dense sand	50

This detailed stratigraphic modeling, including SPT values, is essential for understanding the vertical variation in soil properties and how they influence pile behavior.

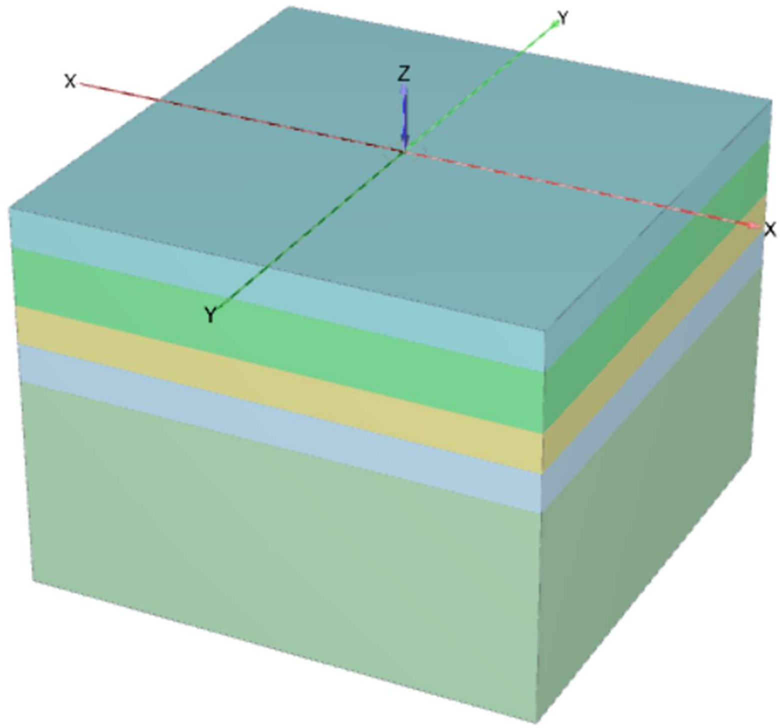

Figure 3-6-1. Detailed Soil Stratigraphy and Pile Modeling in PLAXIS 3D

- ***Soil Properties:*** The relevant soil properties, including unit weight (γ), shear strength parameters (c, φ), and stiffness parameters, will be input into the model. These properties were estimated using empirical equations as mentioned in the Evaluation of the Analytical Methods.
  - **Unit Weight (γ)**: The unit weight of each soil layer, derived from the geotechnical investigation, is essential for realistic simulation. It influences the vertical stress distribution within the soil and around the pile.
  - **Shear Strength Parameters (c, φ)**:
    - **Cohesion (c)**: This parameter is vital for understanding the soil's ability to stick together and resist shear forces.
    - **Angle of Internal Friction (φ)**: This angle represents the shear strength due to friction between soil particles. Both parameters are critical for defining the soil's resistance to shear stresses.

- **Stiffness Parameters**: Stiffness parameters, including the modulus of elasticity (E) and Poisson's ratio (ν) for each soil layer, are essential for predicting how the soil will deform under loading. These parameters help in understanding the soil's response to pile loading.

  - **Poisson’s ration (ν):**

Table 3-5. Poisson's Ratio (ν') for Different Soil Types and Conditions

Soil type	Description	ν'
Clay	Soft	0.35–0.4
	Medium	0.3–0.35
	Stiff	0.2–0.3
Sand	Loose	0.15–0.25
	Medium	0.25–0.3
	Dense	0.25–0.35

- **Modulus of Elasticity (E):**

The modulus of Elasticity (E) was determined by the help of correlation with SPT value (for cohesionless soils) and soil type (cohesive soils) collected from J. E. Bowles, *Foundation Analysis and Design. 1997.*

- For Cohesive Soil

TABLE 2-8
Value range* for the static stress-strain modulus E_s for selected soils (see also Table 5-6)
Field values depend on stress history, water content, density, and age of deposit

Soil	E_s, MPa
Clay	
Very soft	2–15
Soft	5–25
Medium	15–50
Hard	50–100
Sandy	25–250
Glacial till	
Loose	10–150
Dense	150–720
Very dense	500–1440
Loess	15–60
Sand	
Silty	5–20
Loose	10–25
Dense	50–81
Sand and gravel	
Loose	50–150
Dense	100–200
Shale	150–5000
Silt	2–20

*Value range is too large to use an "average" value for design.

Figure 3-7-1. Value range of Modulus of Elasticity for Selected soil

- For Cohesionless Soil:

TABLE 5-6
Equations for stress-strain modulus E_s by several test methods

E_s in kPa for SPT and units of q_c for CPT; divide kPa by 50 to obtain ksf. The N values should be estimated as N_{55} and not N_{70}. Refer also to Tables 2-7 and 2-8.

Soil	SPT	CPT
Sand (normally consolidated)	$E_s = 500(N + 15)$	$E_s = (2 \text{ to } 4)q_u$
	$= 7000\sqrt{N}$	$= 8000\sqrt{q_c}$
	$= 6000N$	— — —
	— — —	$E_s = 1.2(3D_r^2 + 2)q_c$
	‡$E_s = (15\,000 \text{ to } 22\,000) \cdot \ln N$	*$E_s = (1 + D_r^2)q_c$
Sand (saturated)	$E_s = 250(N + 15)$	$E_s = Fq_c$
		$e = 1.0 \quad F = 3.5$
		$e = 0.6 \quad F = 7.0$
Sands, all (norm. consol.)	¶$E_s = (2600 \text{ to } 2900)N$	
Sand (overconsolidated)	†$E_s = 40\,000 + 1050N$	$E_s = (6 \text{ to } 30)q_c$
	$E_{s(\text{OCR})} \approx E_{s,nc}\sqrt{\text{OCR}}$	
Gravelly sand	$E_s = 1200(N + 6)$	
	$= 600(N + 6) \quad N \le 15$	
	$= 600(N + 6) + 2000 \quad N > 15$	
Clayey sand	$E_s = 320(N + 15)$	$E_s = (3 \text{ to } 6)q_c$
Silts, sandy silt, or clayey silt	$E_s = 300(N + 6)$	$E_s = (1 \text{ to } 2)q_c$
	If $q_c < 2500$ kPa use $\quad {}^{\$}E_s' = 2.5q_c$	
	$2500 < q_c < 5000$ use $\quad E_s' = 4q_c + 5000$	
	where	
	$E_s' = \text{constrained modulus} = \dfrac{E_s(1-\mu)}{(1+\mu)(1-2\mu)} = \dfrac{1}{m_v}$	
Soft clay or clayey silt		$E_s = (3 \text{ to } 8)q_c$

Figure 3-8-1. Value range of Modulus of Elasticity for Selected soil Figure

According to the foot-notes; for sand, using E_s = 250 or 500(N +15) may result in a modulus that is conservative and potentially too small. It is recommended to compute E_s using this and additional equations, then average the results for a more accurate estimate.

The selection of appropriate soil constitutive models is crucial for accurately simulating soil behavior under loading conditions. In PLAXIS 3D, two commonly used models are the Mohr-Coulomb model and the Hardening Soil model. The Mohr-Coulomb model is suitable for representing simple, linear elastic-perfectly plastic behavior, while the Hardening Soil model is better suited for capturing complex, nonlinear soil behavior, thereby providing a more realistic simulation of actual soil conditions under various loading scenarios.

For this study, due to simplicity and the lack of a comprehensive data set, we have chosen to simulate the static load test on TP-4 using the Mohr-Coulomb model as the constitutive model in PLAXIS 3D. This approach ensures a straightforward and reliable analysis while still providing valuable insights into the pile-soil interaction.

3.3.4.3 *Meshing*

Meshing is a vital step in finite element analysis (FEA) since it directly influences the degree of detail and accuracy of the simulation results. The mesh consists of discrete elements which divide the computational domain into smaller, manageable chunks. Each element approximates the behavior of the physical system, enabling the analysis of complex structures under different loading conditions.

The study involved doing simulations utilizing all possible meshing options in PLAXIS 3D, ranging from very coarse to very fine. This complete methodology enables a comparative analysis to ascertain the mesh density that yields results that closely correspond with the field test data. The selection of mesh density has to find an acceptable balance between the requirement for accuracy and the requirement for computational efficiency.

- ***Rationale for Different Types of Meshing in Finite Element Analysis***

  In finite element analysis (FEA), the choice of meshing density significantly impacts the accuracy and computational efficiency of the simulations. For this study, various meshing options available in PLAXIS 3D were employed, ranging from very coarse to very fine. Each type of meshing has its own advantages and limitations, and the rationale for choosing each type is explained below, accompanied by relevant images and computation times.

- **Very Coarse Meshing**

  - **Rationale:**

  1. **Initial Exploratory Simulations:**
     - Very coarse meshing is ideal for preliminary simulations aimed at gaining an initial understanding of the system's behavior. It allows for rapid iterations and helps in identifying critical areas that might require more detailed analysis.
  2. **Computational Speed:**
     - This meshing option significantly reduces the number of elements and nodes, thereby decreasing computational time and memory usage. This is particularly useful when computational resources are limited or when multiple preliminary simulations are needed.
  3. **Simplified Geometry Representation:**
     - Very coarse meshing provides a simplified representation of the geometry, which can be sufficient for initial scoping studies or when a rough estimate of the system's response is adequate.

  - **Computation Time:**
     - **4 minutes 50 seconds**

- **Image:**

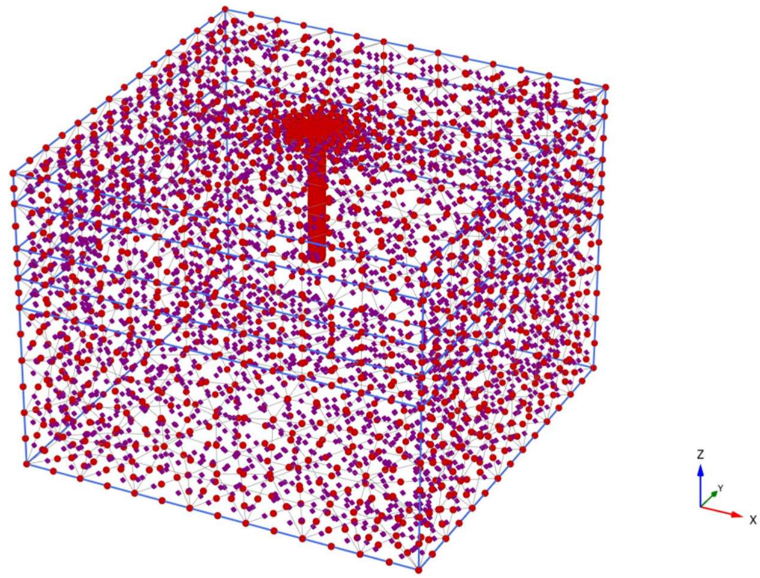

Figure 3-9-1. Very Coarse Mesh Generation in PLAXIS 3D for Pile and Soil Model

- **Coarse Meshing**
  - **Rationale:**

1. **Broad System Behavior Analysis:** Coarse meshing strikes a balance between computational efficiency and capturing the overall behavior of the pile-soil system. It is suitable for studies where a general understanding of the load distribution and deformation patterns is required.
2. **Reduced Computational Cost:** Like very coarse meshing, this option keeps computational costs low while offering a more refined analysis than the very coarse option, making it ideal for initial design iterations.
3. **Identification of Critical Zones:** Coarse meshing helps in pinpointing critical zones that exhibit significant stress concentrations or deformations, which can then be analyzed in more detail with finer meshes.

- **Computation Time:**
  - **4 minutes 47 seconds**
- **Image:**

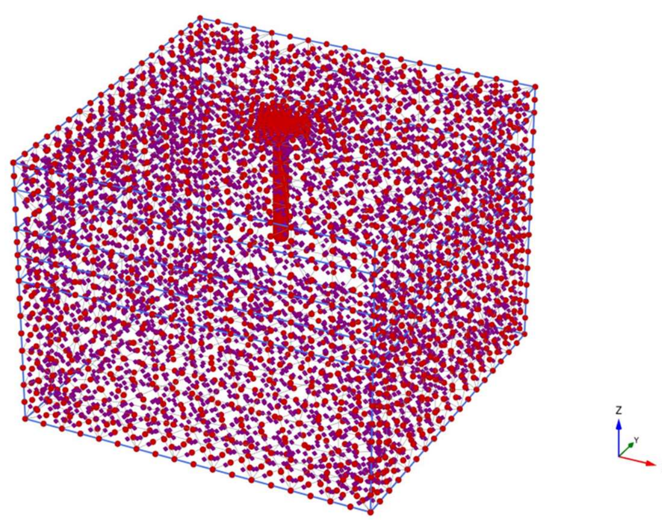

Figure 3-10-1. Coarse Mesh Generation in PLAXIS 3D for Pile and Soil Model

- **Medium Meshing**

  - **Rationale:**

    1. **Detailed Analysis of Key Features:** Medium meshing provides a good compromise between detail and computational efficiency. It captures more geometric details and stress distributions than coarse meshing, making it suitable for more detailed analyses.
    2. **Balanced Accuracy and Efficiency:**
       - This meshing option is often used when there is a need for reasonably accurate results without the extensive computational cost associated with finer meshes.
    3. **Preliminary Validation:**
       - Medium meshing can be used for preliminary validation of the model against empirical data or simplified analytical solutions, providing confidence before moving to finer meshing.

  - **Computation Time:**
    - **4 minutes 44 seconds**

- **Image:**

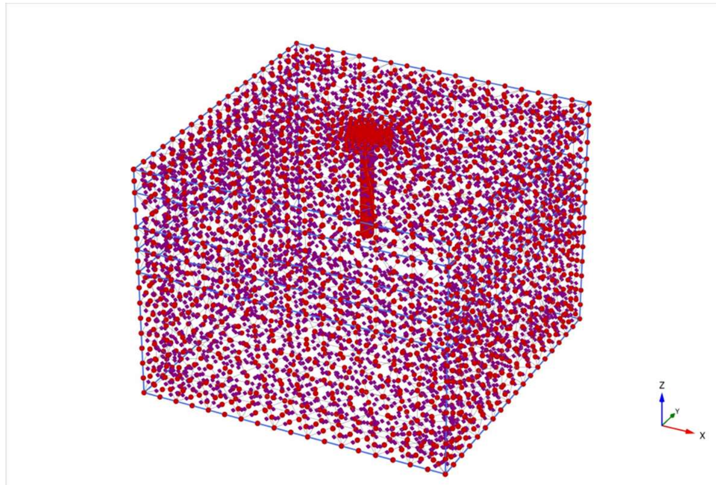

Figure 3-11-1. Medium Mesh Generation in PLAXIS 3D for Pile and Soil Model

- **Fine Meshing**

  - **Rationale:**

1. **High-Accuracy Requirements:**

   - Fine meshing is used when high accuracy is paramount. It captures detailed stress distributions, deformations, and interactions within the pile-soil system, making it suitable for final design validation and detailed analysis.

2. **Complex Geometry Representation:**

   - This meshing option accurately represents complex geometries and small features, which is essential for capturing localized effects and interactions within the model.

3. **Validation Against Field Data:**

   - This meshing option accurately represents complex geometries and small features, which is essential for capturing localized effects and interactions within the model.

- **Computation Time:**

  - **10 minutes 03 seconds**

- **Image:**

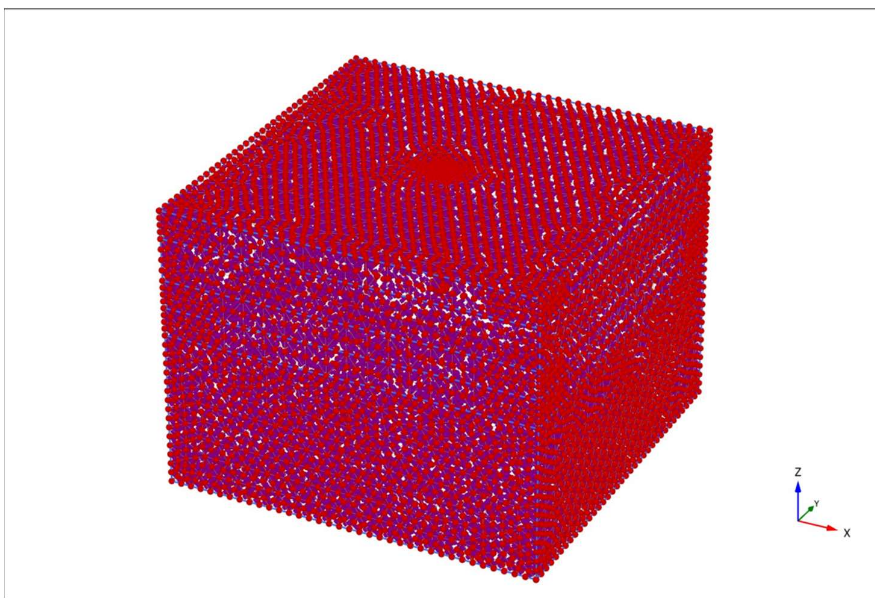

Figure 3-12-1. Fine Mesh Generation in PLAXIS 3D for Pile and Soil Model

- **Very Fine Meshing**
  - **Rationale:**

  1. **Maximum Detail and Precision:**
     - Very fine meshing offers the highest level of detail and precision, capturing minute variations in stress and deformation. It is essential for studies where the utmost accuracy is required.
  2. **Complex Interaction Analysis:**
     - This meshing option is ideal for analyzing complex interactions within the pile-soil system, such as those involving intricate contact conditions or highly localized stress concentrations.
  3. **Benchmarking and Validation:**
     - Very fine meshing is used for benchmarking the FEA model against high-fidelity experimental data or highly detailed analytical models, ensuring the highest confidence in the simulation results.

  - **Computation Time:**
    - **28 minutes 12 seconds**

- **Image:**

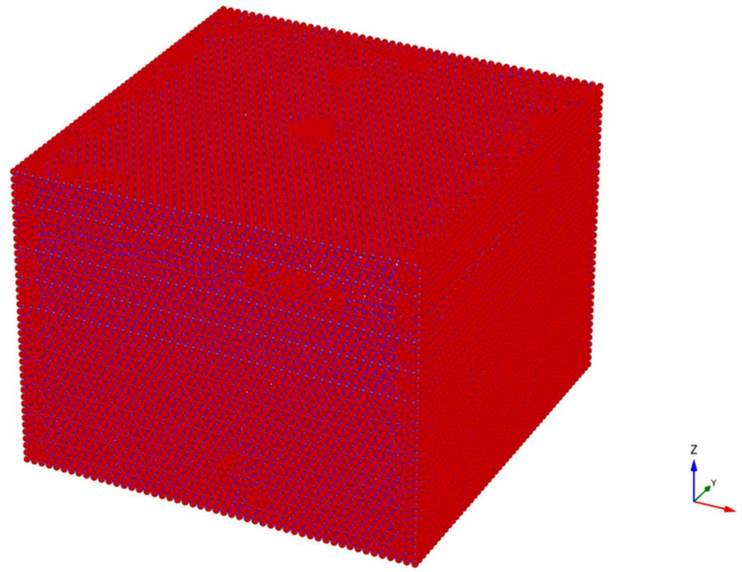

Figure 3-13-1. Very Fine Mesh Generation in PLAXIS 3D for Pile and Soil Model

- ***Meshing Procedure in PLAXIS 3D***

PLAXIS 3D provides robust tools for creating and optimizing meshes. The software allows for automated meshing processes with options for local refinement. In this study, the following steps were undertaken in PLAXIS 3D:

1. **Automatic Mesh Generation:**
   - PLAXIS 3D's automatic meshing tool was utilized to create initial meshes across the range from very coarse to very fine. This tool efficiently generates a mesh based on the defined geometry and desired element size.
2. **Local Refinement:**
   - For each meshing option, local refinement was applied to regions of high interest, such as the pile-soil interface and the pile tip. This ensures critical areas are modeled with higher accuracy without unnecessarily refining the entire domain.

3. **Mesh Optimization:**
   - The generated meshes were reviewed and optimized to ensure element quality. Elements with poor aspect ratios or significant distortions were adjusted to enhance the overall quality and reliability of the mesh.

By employing a range of meshing densities, this study aims to provide a comprehensive understanding of the pile-soil interaction and identify the most effective meshing strategy for accurate and efficient FEA simulations.

3.3.4.4 *Table of Input Parameters:*

To provide a comprehensive overview, the following table summarizes the input parameters for soils and piles used in the PLAXIS 3D model:

Table 3-6. Table of Input Parameters in PLAXIS 3D

Material	Symbol	1st Layer Clay	2nd Layer Clay	3rd Layer Silt	4th Layer Medium Dense Sand	5th Layer Dense Sand	Pile	Unit
Material Model	-	Mohr-Coulomb model	Mohr-Coulomb model	Mohr-Coulomb model	Mohr-Coulomb model	Mohr-Coulomb model	Linear-Plastic	-
Drainage Type		Undrained	Undrained	Undrained	Drained	Drained		
Unit Weight	Υ	17.27	18.85	17.5	22	22.78	25	KN/m^3
Saturated Unit Weight	Υ_{sat}	18.85	20.3	20.5	22.78	22.78	25	KN/m^3
Stiffness Modulus	E	20000	35000	10000	95000	100000	20×10^6	KN/m^2
Poisson's Ratio	ϑ	0.4	0.35	0.2	0.3	0.25	0.16	-
Earth Pressure Co-efficient	K_0	0.9128	0.9128	0.5	0.2867		-	-
Friction Angle	φ	5	5	30	48.5	46	-	-

Material	Symbol	1st Layer Clay	2nd Layer Clay	3rd Layer Silt	4th Layer Medium Dense Sand	5th Layer Dense Sand	Pile	Unit
Cohesion	c_u	62.5	87.5	21	3	1	-	KN/m^2
R_{inter}	R_{inter}	0.7	0.7	0.75	0.8	0.8	1	
Void Ratio	e	0.65	0.6	0.5	0.45	0.43	0.1	

3.3.4.5 *Loading Conditions*

- **Loading Protocol:**

The loading protocol is a critical component of the Static Load Tests (SLTs), and it will be rigorously replicated in the PLAXIS 3D model to ensure the simulation accurately reflects the real-world testing conditions. The specific steps involved in the loading protocol are as follows:

- **Incremental Loading**: The total load was divided into eight increments. Each load increment was applied sequentially, with careful supervision.

- **Load Reduction**: After the maximum load was applied, the load was gradually reduced in four increments. This step helps in understanding the elastic recovery of the pile and any residual deformations.

In the PLAXIS 3D model, this detailed loading protocol will be staunchly represented. The load reduction process was be accurately modeled to replicate the actual static load test (SLT) conditions.

- **Boundary Conditions:** Boundary conditions are vital for accurately simulating the constraints and interactions between the pile and the surrounding soil. In the PLAXIS 3D model, the following boundary conditions will be defined:

  - **Lateral Boundaries**: The lateral boundaries of the soil model will be set far enough from the pile to minimize boundary effects and simulate an infinite soil mass. Typically, lateral boundaries are placed at a distance of at least five times the pile diameter from the pile centerline to ensure they do not influence the results.

- **Bottom Boundary**: The bottom boundary will be set as a fixed boundary (zero displacement) to represent the incompressible bedrock or a very stiff soil layer at depth, ensuring no vertical displacement occurs at this boundary.

- **Top Boundary**: The top boundary will be free to allow vertical displacement, simulating the ground surface where the pile load is applied.

- **Interface Elements**: Interface elements will be deployed to model the interaction between the pile and surrounding soil. The interface components serve a vital part in capturing the shear transfer and any of the slippage between the pile and the soil. A representative image of the pile interface is provided below.

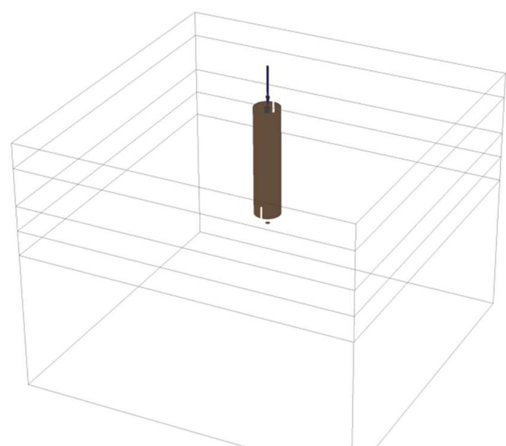

Figure 3-14-1. Pile interface to replicate and capture soil-pile interactions.

By accurately defining these boundary conditions, the PLAXIS 3D model will be capable of exactly replicating the constraints and interactions observed in the real world. The interface properties will ensure that the load transfer mechanisms and interaction effects between the pile and the soil are practically captured, significantly influencing the pile's behavior under load. Accurate boundary conditions are required for obtaining an accurate and dependable simulation, which is required for validating the empirical and analytical approaches employed in calculating the ultimate bearing capacity of piles.

3.3.4.6 *Model Calibration*

The PLAXIS 3D model will be systematically calibrated using the measured data acquired from the static load tests (SLTs). Model calibration is an essential process with the goal of achieving a close correspondence between the simulated results and the pile's actual load-settlement behavior, taking into account various loading conditions. This procedure involves performing iterative adjustments to soil attributes and model parameters using empirical data until an appropriate correlation is established between the predicted and actual performance. The calibration process is enhanced by simulating the model under various meshing circumstances, ranging from coarse to fine, in order to determine the optimal mesh density that yields the highest level of accuracy.

- **Calibration Process**
  - **Initial Parameter Selection**

    1. **Soil Properties**: Initial estimates for soil properties such as unit weight (γ), cohesion (c), angle of internal friction (φ), and stiffness parameters (e.g., Young's modulus E, Poisson's ratio ν) were obtained from the geotechnical investigation report. Empirical correlations and laboratory test results also inform these initial values.
    2. **Pile Properties**: The pile's material properties, such as Young's modulus, Poisson's ratio, and density, were set based on known specifications for the concrete pile used in the SLTs.

  - **Baseline Simulation**

    1. **Load-Settlement Curve Generation**: A baseline simulation was run in PLAXIS 3D using the initial parameters. The load-settlement curve was generated from this simulation serves as the preliminary point for comparison with the SLT results.
    2. **Evaluation**: The simulated load-settlement curve was compared with the measured SLT curve. Differences in key aspects, such as initial stiffness, ultimate bearing capacity, and the overall shape of the curve, were identified.

- **Parameter Adjustment**

1. **Iterative Process**: Parameters were adjusted iteratively to minimize the differences between the simulated and measured curves. Key adjustments typically include:

    - **Cohesion (c)**: Adjusting cohesion to better reflect the soil's shear strength characteristics.
    - **Angle of Internal Friction (φ)**: Fine-tuning the friction angle to match the observed resistance to shear stresses.
    - **Young's Modulus (E)**: Modifying the stiffness to better simulate soil deformation under load.
    - **Pile-Soil Interface Properties**: Adjusting the pile-soil friction angle (δ) to accurately capture load transfer mechanisms.

- **Meshing Condition Analysis**

1. **Mesh Variations**: The model was simulated using different meshing conditions ranging from very coarse to very fine.
2. **Mesh Evaluation**: Each meshing condition was evaluated to determine its impact on the load-settlement behavior. The mesh density that provides the most accurate results with minimal computational cost is identified.

- **Sensitivity Analysis**

1. **Parameter Sensitivity**: A sensitivity analysis was conducted to understand the influence of each parameter on the model's output. Parameters with a significant impact on the load-settlement behavior were identified for closer scrutiny and finer adjustments.
2. **Critical Parameters**: Focus was placed on adjusting parameters that show the highest sensitivity and largest deviations in the initial simulations.

- **Validation**

  1. **Refined Simulation**: A refined simulation was run using the adjusted parameters. The new load-settlement curve was compared with the SLT data to check for improvements in accuracy.
  2. **Quantitative Comparison**: Quantitative comparisons were done by calculating the percentage deviation were calculated to objectively assess the calibration quality.
  3. **Qualitative Comparison**: A qualitative comparison was also performed by visually inspecting the alignment of the simulated and measured curves, focusing on key points like initial stiffness, yield point, and ultimate capacity.

- **Final Calibration**

  1. **Convergence Criteria**: The calibration process was continued iteratively until the differences between the simulated and measured curves fall within an acceptable range. Convergence criteria included both statistical thresholds and visual inspection benchmarks.
  2. **Documentation**: All adjustments and their justifications were thoroughly documented. This includes the rationale for changes in soil and pile properties, as well as the results of sensitivity analyses.

By simulating the model in various meshing conditions, the calibration ensured that the most appropriate mesh density was selected for accurate and efficient simulation results.

3.4 Summary

The method for evaluating the ultimate bearing capacity (UBC) of piles under static incremental loading consists of an extensive strategy focused on well-documented data from static load tests (SLTs). The primary objective is to assess the accuracy and limitations of different approaches, such as analytical methods, empirical correlations, and PLAXIS 3D simulations. The method begins by conducting extensive data gathering, specifically targeting details concerning the setup of the tests, including pile geometry, site conditions, and the specific loading protocols applied throughout the SLTs. The piles' behavior is thoroughly documented, capturing the entire load-settlement curve, which offers vital insights into the piles' response to increasing loads. In addition, pertinent soil data, such as in-situ test results like Standard Penetration Test (SPT) values and laboratory-determined soil parameters, are collected for comprehensive analysis.

The effectiveness of analytical approaches is assessed by utilizing proven mathematical correlations between soil parameters and pile geometry to estimate the UBC. The accuracy of these predictions is evaluated by comparing them with the measured SLT data. Simultaneously, the researchers examine empirical methods that evaluate the relationships between in-situ test data and the observed behavior of piles during SLTs. The empirical UBC estimations are also assessed against the SLT outcomes to determine their efficacy.

A complete model is constructed for PLAXIS 3D simulations using the obtained soil and pile data. This includes defining the soil stratigraphy and properties, such as unit weight, cohesion, friction angle, and stiffness parameters, which are input into the model to accurately reflect the subsurface conditions. The model was simulated for all possible meshing conditions, ranging from very fine to very coarse. Coarse meshing was implemented to strike a balance between computational performance and model accuracy. Initial simulations were conducted using approximated parameters, which were then iteratively adjusted to closely align the simulated and measured load-settlement behavior. A sensitivity analysis was performed to identify the crucial parameters that significantly impact the model's accuracy.

This rigorous technique provided a robust foundation for evaluating various UBC estimation methods, offering valuable insights into their effectiveness and applicability for geotechnical engineering applications. Precise and reliable UBC estimations were aimed at being achieved through meticulous data collection, comprehensive analysis, and thorough model calibration.

CHAPTER 4

Results and Discussion

4.1 Introduction

The focus of this study is to assess and validate the behavior of pile load settlement by implementing finite element analysis (FEA) in PLAXIS 3D. This is achieved by comparing the results from empirical correlations, analytical methods, and field static load tests (SLTs). The primary objective is to determine the most reliable and accurate method for estimating the ultimate bearing capacity (UBC) and creating Q-z curves.

The study has systematically organized the outcomes to achieve its specific goals. Initially, load-settlement data and Q-z curves developed using PLAXIS 3D simulations are compared with those acquired from SLTs. This comparison is performed across different meshing alternatives in PLAXIS 3D to ascertain the optimal balance between accuracy and computational efficiency. Although very fine mesh sizes yield the most accurate results compared to field tests, medium and coarse mesh sizes produce nearly consistent Q-z curves while requiring significantly less computing time.

The subsequent sections emphasize evaluating the ultimate bearing capacity (UBC) using PLAXIS 3D. This evaluation incorporates conservative soil modulus values, estimated based on empirical correlations outlined by Bowles, to determine the modulus of elasticity. These conservative estimates are then compared to the UBC derived from the initial PLAXIS 3D simulation, which utilized non-conservative settings, as well as the estimated UBC obtained from the practical static load test (SLT) dataset. The comparative analysis integrates findings from empirical and analytical approaches, along with SLT data, to offer a comprehensive assessment of each methodology.

The final part of the discussion integrates these findings, highlighting the reliability and practicality of different methods in geotechnical engineering. The strengths and limitations of each approach are discussed, providing valuable insights into their application in real-world scenarios. This comprehensive evaluation aims to inform best practices for estimating pile load settlement and UBC, enhancing the accuracy and efficiency of geotechnical engineering projects.

4.2 Comparison of Load-Settlement Data and Q-z Curves

4.2.1 Introduction:

The load-settlement data and Q-z curves were developed using different meshing alternatives accessible in PLAXIS 3D simulations. Five distinct load-settlement datasets and their related Q-z curves were generated for each type of meshing. The datasets were thoroughly compared with the load-settlement data and Q-z curves acquired from practical Static Load Tests (SLTs) done in the field. The purpose of this comparison was to evaluate the accuracy and reliability of the PLAXIS 3D simulations in regard to actual data from the field test results which considered as the benchmark.

The generated load-settlement data and Q-z curves for each meshing option are provided below for visual representation. These visual comparisons offer a clear understanding of how different meshing options impact the accuracy of the simulations and their correlation with the practical SLT results. By analyzing these illustrations, we can identify the meshing option that most accurately replicates the load-settlement dataset and Q-z curve obtained from practical SLTs. Additionally, by incorporating the iteration and calculation times for different meshing options, this analysis helps determine the optimal mesh size that balances computational efficiency and precision. This ensures that the simulated outcomes closely align with real-world measurements.

Table 4-1-1. Chart of Applied load and its corresponding settlement data for (I) very coarse mesh, (II) coarse mesh, (III) Medium mesh, (IV) Fine mesh, (V) Very Fine mesh & (VI) Field Test

Applied Load (Ton)	Very Coarse	Coarse	Medium	Fine	Very Fine	Field test
0	-0.07741	-0.0792	-0.0732	-0.0668	-0.0496	0
24.99	-0.9231	-0.8228	-0.7839	-0.7019	-0.6732	-0.462
49.98	-1.7012	-1.7012	-1.6187	-1.4422	-1.3982	-1.154
74.97	-3.3274	-2.8215	-2.7036	-2.3881	-2.3117	-2.154
99.96	-4.8109	-4.0532	-3.8693	-3.4334	-3.3159	-3.077
124.95	-6.3622	-5.3752	-5.1129	-4.5383	-4.4099	-4.154
149.94	-8.2050	-6.9135	-6.6299	-5.9164	-6.1374	-5.307
174.93	-13.052	-11.275	-11.191	-10.134	-11.394	-9.007
187.6749	-16.5304	-14.496	-13.941	-12.889	-14.323	-14.385

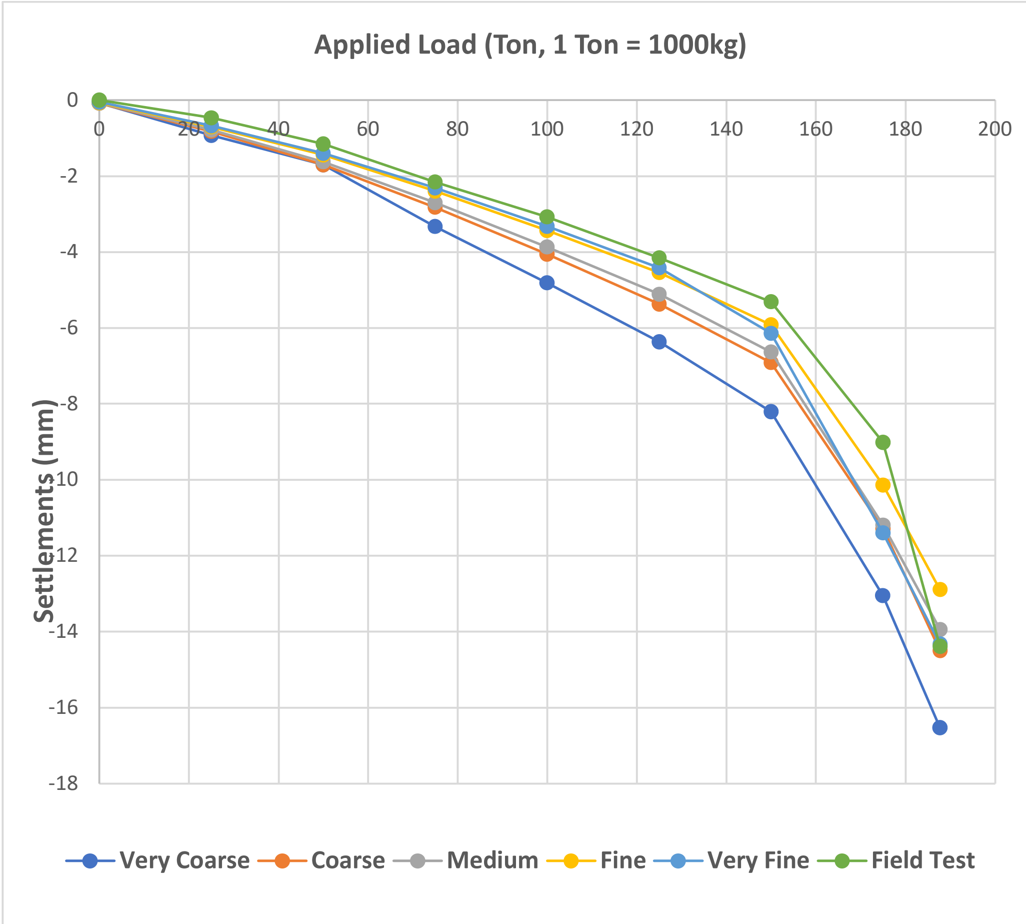

Figure 4-1-1. Effect of mesh size on model prediction

Table 4-2-2. Mesh size and its corresponding elapsed time

Granulometry	Elapsed Time
Very Coarse	4 min 50 sec
Coarse	4 min 47 sec
Medium	4 min 44 sec
Fine	10 min 03 sec
Very Fine	28 min 12 sec

4.2.2 Discussion

- **Very Fine Mesh**
  - **Accuracy**: Provides the closest match to the Q-z curve from the SLT, making it ideal for detailed and precise simulations.
  - **Computational Time**: Requires significantly more time at 28 minutes and 12 seconds, which can be a limiting factor in scenarios requiring multiple simulations or when computational resources are constrained.
- **Fine Mesh**
  - **Accuracy**: Delivers very high accuracy, closely matching the SLT Q-z curve results.
  - **Computational Time**: Offers a reasonable trade-off between accuracy and computational time, taking 10 minutes and 03 seconds.
- **Medium Mesh**
  - **Accuracy**: Produces very high accuracy, with Q-z curves nearly identical to the SLT results.
  - **Computational Time**: Significantly shorter at 4 minutes and 44 seconds, making it highly efficient for practical applications.
- **Coarse Mesh**
  - **Accuracy**: Provides high accuracy, making it a reliable choice for producing Q-z curves that are close to the SLT results.
  - **Computational Time**: Comparable to the medium mesh, with a calculation time of 4 minutes and 47 seconds.

4.2.3 Recommendation:

In this study, our key objective is to achieve the maximum level of accuracy in simulating load-settlement behavior. Therefore, the very fine mesh option is the perfect choice here, as it provides the closest match to the practical SLT results. However, for practical applications, it is advisable to use medium or coarse meshing options. These mesh sizes yield nearly identical Q-z curves to the SLT results while significantly reducing calculation times compared to the very fine mesh. This balance ensures that simulations are both accurate and efficient, making them more practical and appropriate for geotechnical engineering projects.

4.3 Estimation of Ultimate Bearing Capacity (UBC)

4.3.1 Introduction

In geotechnical engineering, accurately estimating the ultimate bearing capacity (UBC) of a pile foundation is critical for ensuring the stability and safety of structures. The ultimate bearing capacity (UBC) determines the maximum load a pile can bear before failure occurs, and thus plays a vital role in the design and analysis of foundation systems. This section focuses on the estimation of ultimate bearing capacity (UBC) using PLAXIS 3D simulation, leveraging the very fine mesh option, which was identified as providing the highest degree of resemblance to practical static load test (SLT) results.

4.3.2 Preferred Meshing Option

The very fine mesh in PLAXIS 3D was selected as the preferred meshing option due to its superior accuracy in simulating the load-settlement behavior of piles. This meshing option captures the intricate details of pile-soil interactions, which are crucial for a precise analysis. Despite its higher computational demand, the very fine mesh offers a detailed and reliable representation of the physical phenomena, making it indispensable for critical evaluations.

- **Load-Settlement Data and Q-z Curve Generation**
  - **Simulation Setup:**
    - PLAXIS 3D was configured with the very fine mesh to simulate the pile load tests. This involved defining the geometry, material properties, and boundary conditions of the pile and surrounding soil.
    - The simulation parameters were carefully chosen to reflect realistic field conditions, ensuring that the generated data would be comparable to actual SLT results.
  - **Data Generation:**
    - The load was applied incrementally to the pile, and the resulting settlements were recorded. This process produced a detailed load-settlement dataset.
    - From this dataset, Q-z curves were derived, illustrating the relationship between the applied load (Q) and the corresponding settlement (z) at various points along the pile.
    - The Q-z curve obtained from the PLAXIS 3D simulation dataset is presented below:

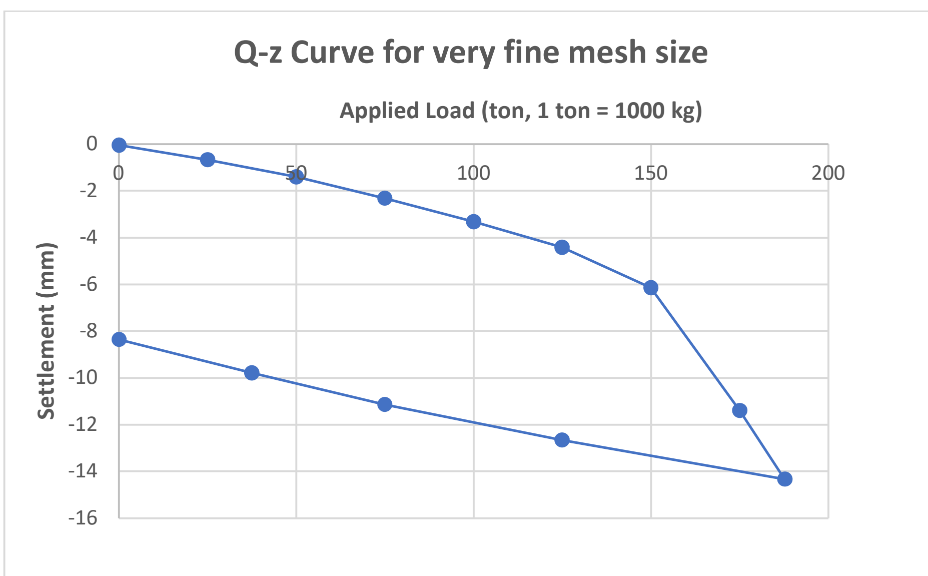

Figure 4-2-1. Q-z Curve for very fine mesh size from PLAXIS 3D

- **Application of the Davisson Method**

To estimate the UBC, the Q-z curve obtained from the very fine mesh simulation was analyzed using the Davisson method (1972), a widely accepted approach in geotechnical engineering.

- **Elastic Deflection Calculation:**

  The estimation of the ultimate bearing capacity (UBC) requires an accurate representation of the pile's elastic deflection, which was drawn alongside the Q-z curve. The elastic deflection is crucial for understanding the pile's initial response to the applied loads.

  The elastic deflection of the pile (Δ) was calculated using the following formula:

$$\Delta = \frac{P \times L}{A \times E}$$

  where:

  — P is the applied axial load,

  — L is the length of the pile,

  — A is the cross-sectional area of the pile,

  — E is the modulus of elasticity of the pile material.

- **Offset Line Calculation:**

  - An offset line was drawn parallel to the elastic deflection. The offset distance was calculated using the formula:

$$offset = 0.15 + 0.1d$$

    where d is the pile diameter. This offset accounts for both the pile diameter and a fixed settlement criterion, ensuring a conservative estimate of UBC.

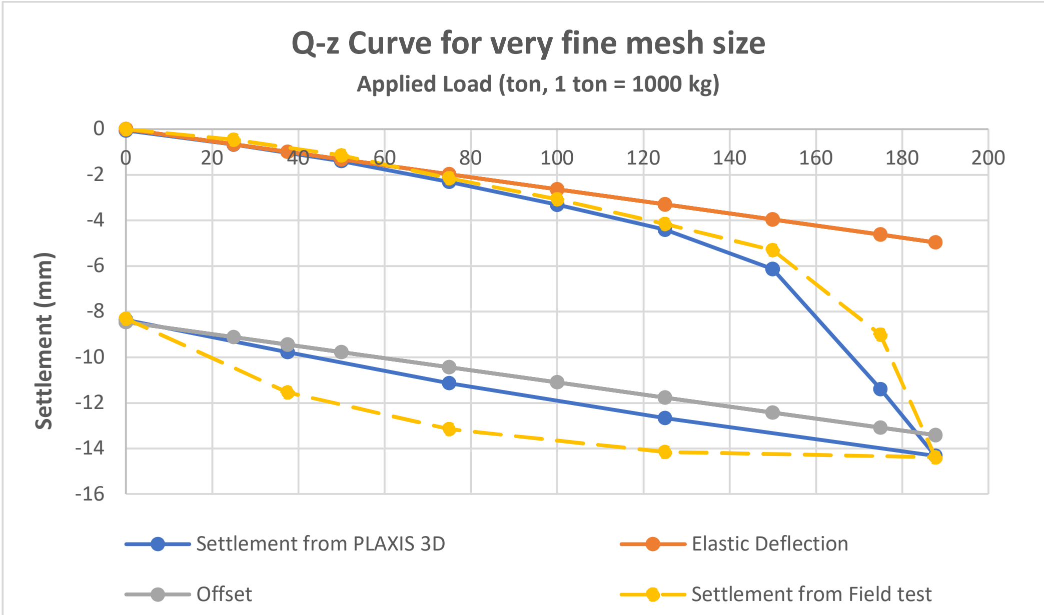

Figure 4-3-1. Q-z Curve for very fine mesh size along with elastic deflection and corresponding off-set line

- **Intersection Point Identification:**
    - The intersection point between the Q-z curve and the offset line was determined. This point signifies the ultimate load capacity of the pile, as per the Davisson method.
- **UBC Estimation:**
    - The load corresponding to the intersection point was recorded as the ultimate bearing capacity. For the very fine mesh simulation, the UBC was found to be 183 tons.

- **Comparison with SLT Results**

The UBC estimated from the very fine mesh simulation was then compared to the UBC derived from actual SLT data:

- **SLT UBC:**
  - The UBC obtained from the practical SLT was 186 tons.
- **PLAXIS 3D UBC:**
  - The UBC from the very fine mesh Q-z curve was 183 tons.
- **Percentage of Deviation:**
  - $\boldsymbol{Deviation} = \frac{\boldsymbol{SLT\ UBC - PLAXIS\ 3D\ UBC}}{\boldsymbol{SLT\ UBC}} \times \mathbf{100}$
  - $\boldsymbol{Deviation} = \frac{\mathbf{186-183}}{\mathbf{186}} \times \mathbf{100}$
  - $\boldsymbol{Deviation} = \mathbf{1.613}\%$

The close agreement between the UBC values obtained from PLAXIS 3D simulations (183 tons) and the SLT (186 tons) demonstrates the accuracy and reliability of the finite element analysis when using a very fine mesh. The percentage deviation of 1.613% indicates that the estimated ultimate bearing capacity closely matches the benchmark provided by the SLT. This validation is crucial as it confirms that the simulation results accurately represent real-world behavior, reinforcing the effectiveness of the PLAXIS 3D modeling approach.

4.4 Estimation of Ultimate Bearing Capacity (UBC) using conservative approach

4.4.1 Introduction:

In this section, the Ultimate Bearing Capacity (UBC) was determined using a conservative approach to model the PLAXIS 3D simulation. This method was chosen to ensure the most reliable and safe estimation of the UBC, taking into account the inherent variability and uncertainties in soil properties and pile behavior. By employing a conservative approach, the study aims to provide a thorough understanding of the lower-bound estimate of the UBC, which is crucial for the design and safety assessment of pile foundations.

This conservative modeling involves using soil modulus values derived from well-established empirical correlations, specifically those outlined by Bowles. These values are intentionally chosen to be on the lower end of the spectrum to avoid overestimating the soil's capacity to support loads. By simulating the load-settlement behavior using these conservative parameters, the study ensures that the resulting UBC is a safe and cautious estimate, reflecting a worst-case scenario that can be relied upon in real-world applications.

Additionally, the conservative approach in PLAXIS 3D simulation helps in identifying the safest and most conservative UBC, which is essential for geotechnical engineering projects where safety is paramount. This method provides engineers and designers with a robust basis for decision-making, ensuring that the pile foundations designed using these estimates will perform safely under the expected loads, even under the most adverse conditions. This comprehensive and cautious estimation process contributes significantly to the overall reliability and safety of geotechnical designs.

4.4.2 Methodology

- **Selection of Preferred Meshing Option**

After identifying the meshing option in PLAXIS 3D that offers the highest degree of resemblance in the load-settlement dataset and Q-z curve with those acquired from practical Static Load Test (SLT) datasets, the very fine mesh was selected as the preferred option. This choice was made despite its higher computational demands due to its superior accuracy in replicating field conditions.

- **Conservative Soil Modulus Values**

The conservative approach involved using soil modulus values derived from empirical correlations outlined by Bowles. These conservative values ensure that the simulations do not overestimate the soil's stiffness, providing a more reliable and cautious estimate of the UBC. The modulus of elasticity (E) for the soil layers, particularly the sand layers, was estimated using the empirical correlation:

$$E = 250 \times (N + 15)\ OR\ 500 \times (N + 15)$$

where N is the Standard Penetration Test (SPT) N-value. These values account for potential variations in soil properties and ensure a conservative estimate of the soil's response under load.

- **Simulation and Analysis**

The load-settlement data and Q-z curves were generated using PLAXIS 3D simulations with the selected very fine mesh and the conservative soil modulus values. The Davisson method (1972) was applied to these Q-z curves to estimate the UBC. This method involves plotting the elastic deflection of the pile alongside the Q-z curve and drawing an offset line parallel to the elastic deflection. The intersection of this offset line with the Q-z curve indicates the UBC.

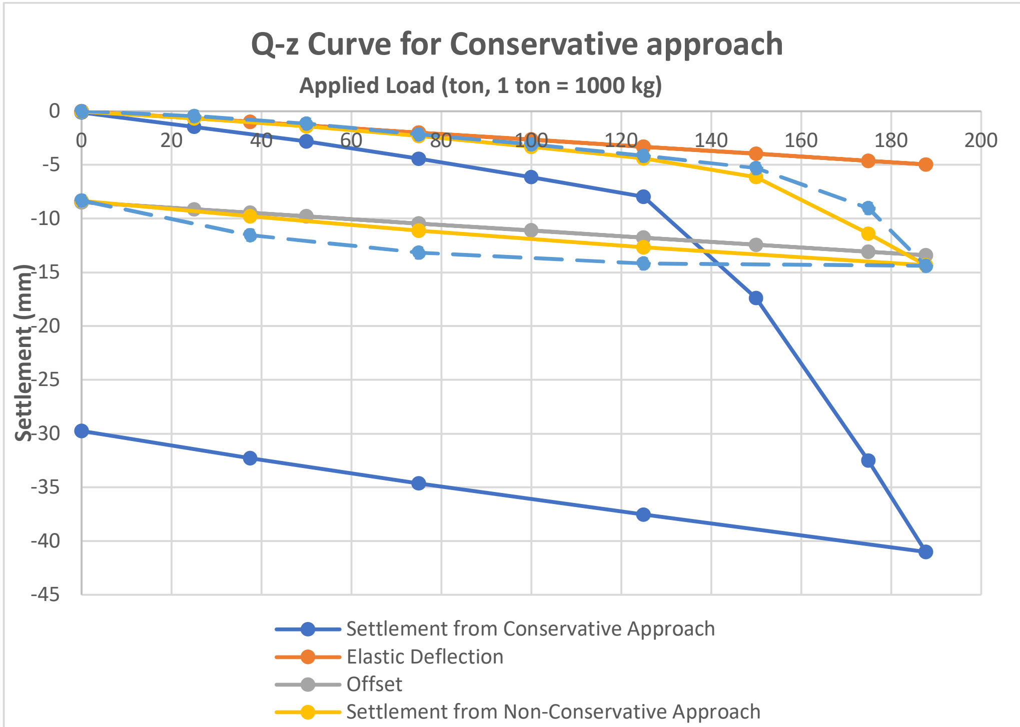

Figure 4-4-1. Q-z Curve for very fine mesh size and conservative approach

- **Results**

The ultimate bearing capacity (UBC) estimated from the Q-z curve obtained using the very fine mesh and conservative soil modulus values in PLAXIS 3D simulations was found to be 136 tons. In contrast, the UBC estimated from the practical Static Load Test (SLT) was 186 tons. While there is a significant deviation between these results, the primary objective of this analysis is not to achieve perfect accuracy in UBC estimation but rather to ensure a conservative and safe estimation of UBC using PLAXIS 3D simulations. The use of a very fine mesh in the simulations aims to closely replicate field conditions, but it is acknowledged that this approach can be computationally intensive. However, the focus remains on providing a conservative estimate that ensures the safety and reliability of the design, recognizing that slightly significant deviations from practical results are acceptable within the context of ensuring overall stability and performance of the pile-soil system. This conservative approach helps to account for various uncertainties in soil properties and modeling assumptions, ultimately contributing to the robustness of geotechnical designs.

- **Percentage of Deviation:**
  - $\boldsymbol{Deviation} = \frac{\boldsymbol{SLT\ UBC - PLAXIS\ 3D\ UBC}}{\boldsymbol{SLT\ UBC}} \times \mathbf{100}$
  - $\boldsymbol{Deviation} = \frac{\mathbf{186 - 136}}{\mathbf{186}} \times \mathbf{100}$
  - $\boldsymbol{Deviation} = \mathbf{26.9}\%$

4.4.3 Discussion

The conservative estimation of UBC using PLAXIS 3D with very fine mesh and empirical soil modulus values provides several advantages:

- **Accuracy:** The use of a very fine mesh ensures high-resolution simulations by dividing the entire model into very small mesh elements. This rigorous approach enhances the precision of the calculations, resulting in highly accurate estimations of the ultimate bearing capacity. By increasing the level of detail in the mesh, the simulation can more closely replicate real-world conditions, leading to optimal accuracy in the results.
- **Safety:** Employing conservative soil modulus values prevents overestimation of soil stiffness, ensuring that the UBC estimates are reliable and safe for design purposes.
- **Validation:** Despite a noticeable deviation of 26.9% between the PLAXIS 3D results and the SLT data, this deviation is acceptable given the conservative approach taken for safety. A deviation under 30% is considered satisfactory in this context. Therefore, the modeling is deemed adequate, with the estimated UBC from PLAXIS 3D being around 27% lower than the benchmark, ensuring a conservative and safe estimation.

4.5 Comparison of UBC Estimation Methods

4.5.1 Introduction

Three different methods were employed to estimate the Ultimate Bearing Capacity (UBC) of piles, and their results were compared with the UBC obtained from Static Load Tests (SLT) conducted on the same site. The primary objective here was to evaluate the accuracy and reliability of each method in predicting the UBC and to provide insights into their practical applicability in geotechnical engineering.

The following chart provides a comparative illustration of the estimated ultimate bearing capacity (UBC) obtained from various methods.

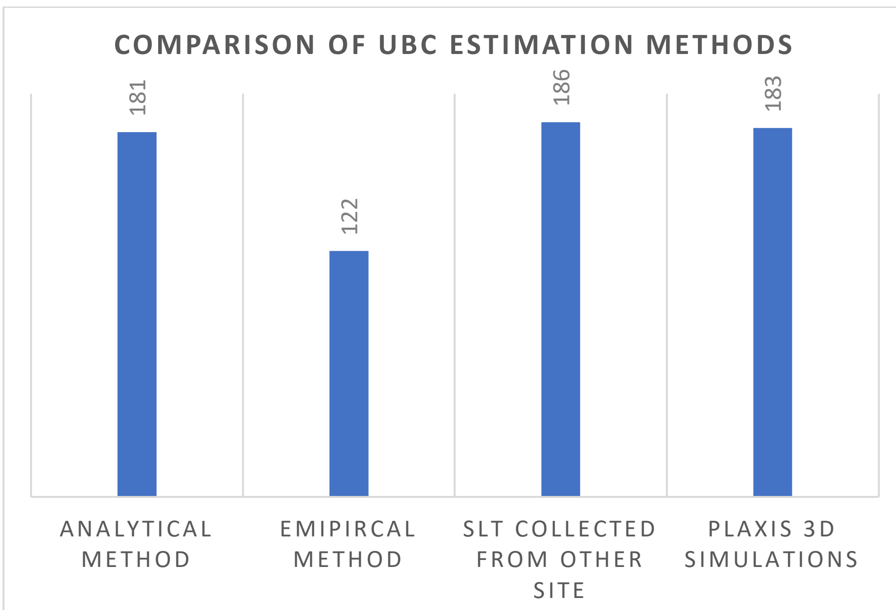

Figure 4-5-1. Comparison between estimated Ultimate Bearing Capacity by using various Methods

- **PLAXIS 3D Simulation:** The PLAXIS 3D simulation, utilizing a very fine mesh, was employed to estimate the UBC. This method involved detailed finite element analysis (FEA) of the pile-soil interaction, capturing the load-settlement behavior accurately. The simulation results yielded a UBC of 183 tons. This value closely aligns with the UBC obtained from the SLT, which was 186 tons. The close correspondence between the PLAXIS 3D simulation and the SLT results underscores the capability of the FEA method in providing accurate predictions of pile behavior under load. The very fine mesh used in PLAXIS 3D ensured a high level of detail and precision, although it required significant computational resources and time.

- **Analytical Method:** The analytical method for estimating UBC is based on classical geotechnical engineering principles, which involve calculations derived from soil mechanics theories and empirical correlations. This method is commonly used due to its simplicity and ease of application. However, the analytical approach estimated the UBC to be 537 tons, which is substantially higher than the SLT result. This significant discrepancy indicates that the analytical method may overestimate the bearing capacity in certain conditions, possibly due to assumptions and simplifications inherent in the method. While the analytical approach provides a quick estimation, it may not always capture the complexities of actual pile-soil interactions.
- **Empirical Method:** The empirical method involves using established empirical correlations and formulas to estimate the UBC based on site-specific soil properties and pile dimensions. Similar to the analytical method, the empirical approach estimated the UBC to be 536 tons. This value is also significantly higher than the UBC obtained from the SLT. The empirical method, while useful for preliminary design and assessment, may not account for all variables influencing pile performance in real-world scenarios. The overestimation observed in both the empirical and analytical methods suggest the need for cautious application and potential calibration against field data.

4.5.2 Comparison with SLT Results:

The SLT conducted on-site provided a UBC of 186 tons, serving as a benchmark for evaluating the other estimation methods. The close alignment of the PLAXIS 3D simulation result (183 tons) with the SLT value confirms the reliability of the finite element analysis approach in accurately predicting pile capacity. In contrast, the analytical and empirical methods, with UBC values of 537 tons and 536 tons respectively, significantly overestimate the capacity, highlighting the limitations of these methods in certain contexts.

4.6 Summary

This chapter focused on the comparative analysis of Ultimate Bearing Capacity (UBC) estimation methods for pile foundations. Using PLAXIS 3D simulations, both standard and conservative approaches were employed to model load-settlement behavior and Q-z curves, which were compared with analytical, empirical methods, and practical Static Load Tests (SLTs). The results highlighted that while PLAXIS 3D simulations provide accurate UBC estimates closely aligned with SLT data, analytical and empirical methods tend to overestimate UBC values. This underscores the importance of integrating numerical simulations with empirical field testing to optimize foundation design decisions in geotechnical engineering applications.

CHAPTER 5
Conclusions and Future Works

5.1 Conclusions

This chapter represents the final outcome of this research undertaking, which sought to assess and validate the behavior of pile foundations through the use of sophisticated numerical simulations and a comparative analysis involving empirical and analytical methodologies. The main objective was to estimate the Ultimate Bearing Capacity (UBC) using PLAXIS 3D simulations, by applying different mesh sizes and conservative methods. The key findings and findings of this study are outlined as follows:

5.1.1 Simulation Accuracy and Meshing Strategies

Extensive simulations conducted in PLAXIS 3D have confirmed that employing a very fine meshing method yields the most accurate representation of load-settlement data and Q-z curves in practical Static Load Test (SLT) outcomes. The use of very fine meshing in this context enables an accurate representation of soil-structure interactions, resulting in improved accuracy of ultimate bearing capacity (UBC) predictions. Nevertheless, medium and coarse mesh sizes also exhibited adequate accuracy while requiring much less processing power. The findings indicate that although using a very fine mesh is optimal for achieving accuracy, using medium and coarse meshes provides a feasible compromise between accuracy and efficiency for typical geotechnical assessments.

- The UBC estimated from PLAXIS 3D simulations using a very fine mesh was found to be 183 tons, closely aligning with the SLT result of 186 tons. This close agreement indicates the reliability of numerical simulations in predicting practical performance and validates the use of PLAXIS 3D for such analyses.
- Comparative analysis with analytical and empirical methods revealed significant discrepancies, with the analytical and empirical UBC estimates exceeding 500 tons. This underscores the conservative nature of numerical simulations when appropriate parameters and boundary conditions are applied. The stark contrast between these methods highlights the need for cautious interpretation of analytical and empirical results and reinforces the value of numerical simulations for more realistic and reliable UBC estimation.

5.1.2 Implications for Geotechnical Engineering Practice

The study underscores the significance of integrating complex numerical modelling with practical field testing to enhance foundation design. This research offers valuable information on load-deformation characteristics and ultimate bearing capacity (UBC) estimation, enabling engineers to make well-informed decisions that achieve a balance between accuracy and computational efficiency. The results strongly support the adoption of PLAXIS 3D simulations as a standard procedure in geotechnical engineering to improve the dependability of designs and the accuracy of performance predictions.

5.1.3 Broader Implications and Applications

In addition to estimating the ultimate bearing capacity (UBC), the results of this study contribute to the progress of methodologies in geotechnical engineering. It highlights the importance of simulation tools such as PLAXIS 3D to enhance the reliability of designs and estimate performance. This research highlights the potential of numerical simulations in enhancing the comprehension of the intricate interactions between soil and structures and in facilitating the development of more efficient and safer foundation systems.

This chapter presents the overall conclusion and recommendation for future works. At first briefly mention what you've done in the research and what is the major contribution. This should present a summary of contributions with focus to achievements, new observations, interpretations and broader implications of results.

The major findings and novel contributions of the research should be mentioned. The major findings should be numbered. Often, these findings are numbered as per the specific aims of the research as mentioned in Chapter 1.

5.2 Limitations and Recommendations for Future Works

Although this study accomplished significant successes, it also highlighted a few limitations and opportunities for future investigation:

5.2.1 Limitations

- **Model Assumptions:** The accuracy of PLAXIS 3D simulations is significantly affected by the input parameters and assumptions established during the modelling process. Improving the characterization of soil parameters and the interaction between piles and soil could enhance the accuracy of simulations.
- **Boundary Conditions:** Further refinement in defining boundary conditions, such as pore water pressures and interface behavior, could yield more precise simulations reflective of field conditions. Accurate representation of these factors is crucial for realistic modeling.
- **Validation:** Although efforts were made to validate the simulation using static load test (SLT) data, extending the validation dataset to include a wider range of soil types and environmental variables would improve the reliability of the simulation results. Implementing a more comprehensive validation approach would improve the ability for adapting the findings to a wider range of situations.

5.2.2 Recommendations for Future Works

- **Parametric Studies:** Conducting parametric studies to investigate the impact of various soil characteristics, pile arrangements, and loading conditions on ultimate bearing capacity (UBC) estimate would yield more profound comprehension of system behavior. These studies can assist in identifying crucial aspects that influence performance and provide guidance for making design enhancements.

- **Long-Term Monitoring:** By implementing a long-term monitoring system for pile foundations, which are exposed to variable loads and environmental conditions, it would be possible to validate simulation predictions over longer durations. This method would provide significant information concerning the durability and performance of foundations under actual conditions.
- **Advanced Modeling Techniques:** Exploring advanced modeling techniques, such as coupled consolidation and thermal effects, could broaden the scope of application for numerical simulations in geotechnical practice. Incorporating these factors would enhance the accuracy of predictions under complex conditions.
- **Integration of Machine Learning:** By using machine learning algorithms, the process of selecting parameters and calibration models based on historical data can be improved, resulting in enhanced predicting abilities and reduced uncertainties. Machine learning can enhance pattern recognition and optimise simulation performance.

Overall, this research improves the comprehension of pile foundation behaviour u sing advanced numerical simulations and emphasises the need for continuous research to improve approaches and address practical challenges in geotechnical engineering. By implementing these suggestions, future research can improve the reliability and us efulness of numerical simulations in real-world engineering applications.
This work enhances geotechnical procedures, leading to safer and more efficient foun dation designs.

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